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CyberSecurity

3 - “Basic Cryptographic Concepts” *[Chapter 2]*

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Outline



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- **Scope** of Cryptography
- **Basic** Operations
- **Cipher** and **Key**
- Example: **Caesar Cipher**
- **Symmetric** Key Encryption
- Modes of Operation
- **Stream** Encryption
- **Secure Hash** Functions
- **Public-Key** Encryption
- Digital Signatures
- Public-Key **Certificates**
- **Random Numbers** Generation
- **Post Quantum** Encryption (NOT IN THE BOOK: study using slides)



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Cryptography (intro)

Scope of Cryptography

Different Cybersecurity Properties



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Different encryption tool will provide different properties (not all, only a subset)

The usage of **cryptographic techniques** is necessary for:

Most common property when we use Cryptography

- **Confidentiality**: “*only the right people **can read** the information*”
- **Authentication**: “*the ability to verify the **sender's identity***” (example, email can be forged: so no Auth property)
- **Integrity**: “*protecting information from **being modified** by **unauthorized parties***” (example, email content can be modified: so no Integrity property)
- **Anonymity**: “*ensuring the **users' anonymity** while communicating*”

Basic Operations



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- **Encryption**: the process of converting **plaintext** into **ciphertext**
- **Decryption**: the process of converting **ciphertext** into **plaintext**

Basic Operations



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- **Encryption**: the process of converting **plaintext** into **ciphertext**
- **Decryption**: the process of converting **ciphertext** into **plaintext**

Can be read

Basic Operations



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- **Encryption**: the process of converting **plaintext** into **ciphertext**
- **Decryption**: the process of converting **ciphertext** into **plaintext**

Can not be read

(not readable by attacker: for example, in a MITM attack, if the packets are encrypted the attacker cannot read it)

Basic Operations



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- **Encryption**: the process of converting **plaintext** into **ciphertext**
- **Decryption**: the process of converting **ciphertext** into **plaintext**

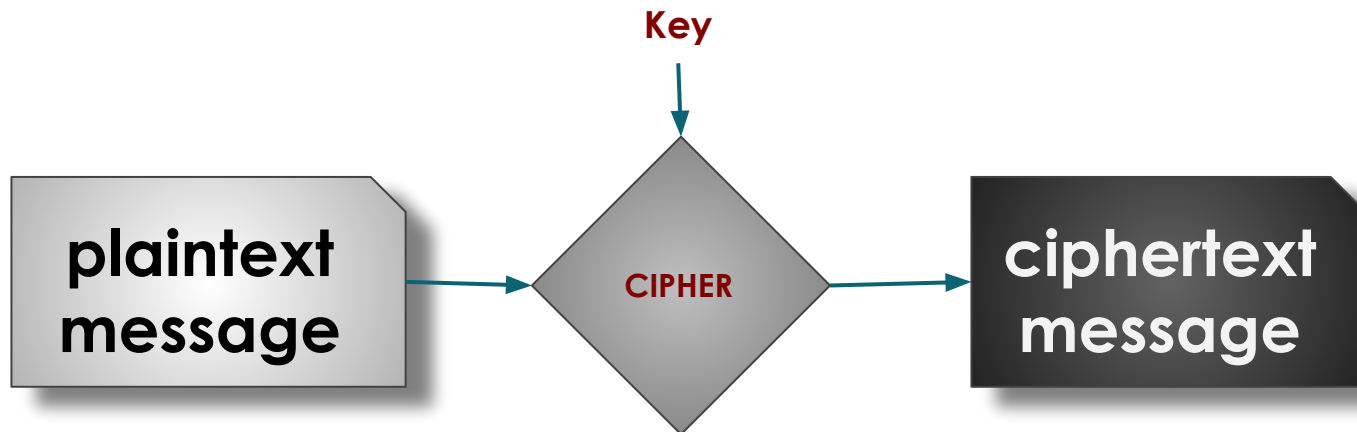


Cipher and Key



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- **Cipher**: an **algorithm** used to **encrypt** and **decrypt** data (must be secure and efficient)
- **Key**: a **piece of information** used by the cipher to encrypt the plaintext (and that is **necessary to decrypt the ciphertext**)



Example: Caesar Cipher



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- A very old example of cipher (i.e. Gaius Julius Caesar) Only letters are encrypted (not numbers and special character like '!' and '?')
- It is a substitution cipher in which each character in the message is substituted by another character
- **For example:** “*substitute the letters in the second row for the letters in the top row to encrypt a message*”

A	B	C	<u>D</u>	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z		
$\downarrow +3$																											
<u>D</u>	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C		

Example: Caesar Cipher



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The letters in the second
row are shifted of a given
number of positions

er (i.e. Gaius Julius Caesar)

which **each character in the message is**
character

the letters in the second row for the letters in
the first row to encrypt a message"

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C

Example: Caesar Cipher



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The letters in the second row are **shifted** of a **given number of positions**

In this example the letters are shifted of 3 positions

...p row to encrypt a message

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C

Example: Caesar Cipher



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The letters in the second row are shifted of a given number of positions

In this example the letters are shifted of 3 positions
("3" is the secret key!)

...p row to encrypt a message

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C

The "Big Enough" threshold in modern cryptography is generally considered to be 2^{128} or higher to resist brute-force attacks from modern hardware. The Caesar Cipher's keyspace doesn't just fall short; it's practically non-existent by comparison.

- Must be BIG ENOUGH to be difficult to evaluate all keys to find the correct one from the POV of the attacker (BIG ENOUGH depends on Computer power)

possible in the

$n-1$ if n is the number of letters in the Alphabet

A B n-1 if n is the number of letters in the Alphabet M N O P Q R S T U V W X Y Z

D E F G H I J K L M N O P Q R S T U V W X Y Z A B C

Example: Caesar Cipher



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- A very old example of cipher (i.e. Julius Caesar)
- It is a simple cipher which **each character in the message is**

How many keys are

possible in the

Caesar Cipher?

Is it secure?

No because it has a short number of keys

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

D E F G H I J K L M N O P Q R S T U V W X Y Z A B C

Example: Caesar Cipher



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- A very old example of cipher (i.e. Julius Caesar)
- It is a simple cipher which each character in the message is

How many keys are possible in the Caesar Cipher?

Is it secure?

Let's play with it!

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z



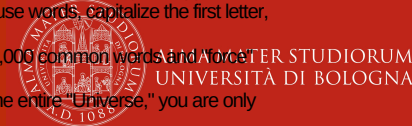
There is a third type of attack that is the Combination of Brute Force and Cryptanalytic Attack. If a cipher has a massive universe of keys, Brute Force is impossible. However, if Cryptanalysis can reveal even a tiny bit of information about the key, the math changes instantly (Reducing the Search Space).

Caesar Cipher: Attacks

Example: Dictionary & Rule-Based Attacks

When attacking passwords, hackers don't just try random characters (aaaa, aaab, aaac). That is pure Brute Force and it's too slow. Instead, they use a hybrid approach:

- Cryptanalysis (Pattern Study): They know humans use words, capitalize the first letter, and put "123" at the end.
- Brute Force (The Execution): They take a list of 100,000 common words and ~~only~~ only the variations that fit the human patterns.
- Why this is powerful: You are no longer searching the entire "Universe," you are only searching the "Likely Universe."



(The two types of attack are used to attack all the types of ciphers, not only the Caesar one)

How is possible to **attack** the Caesar Cipher? (i.e. **to get the cleartext**

without having the secret key)

It is normal that if i steal the key i can get the clear text (we are attacking the Cipher, not the device). If we get the key, is not a failure of the Cipher, but a failure of the security of the device

In Modern Ciphers, is difficult to do Brute Force Attacks (universe of keys and time required are too large)

Brute Force attack: **we can try all the possible keys**

- ^{fattibile} **is it feasible?** It depends on the **number of keys** (i.e. **key size**) and the **time required**

if the Brute Force attack is difficult to do or costly, we do this

to check each key (... *computers are fast!*)

The system designer selects an algorithm with a sufficiently large key space so that, given current computing power and its expected future development, the time required to test every key renders the attack impractical for the entire period during which the information must remain confidential

Cryptanalytic attack: (in this case) **we can use the frequency of characters** (or other **weaknesses** of think kind)

that are in the Design of the Cipher

- **in human languages each letter has a different frequency of use**

the Caesar Cipher translates letters and therefore it translates also their frequency

The Caesar Cipher provides zero confusion. It hides the letters, but it leaks the pattern of the language perfectly. This means an attacker doesn't even need to try all the keys

For example, in English the letter E is the most used. So, if we see that a letter of the cipher text has the same frequency, that letter can be the E (so, we can find the key from the pattern of the language)

This is an example of how you can exploit vulnerabilities in the Design of the Cipher

50% is not the probability of guessing correctly on the first try, but the probability that the key lies in the first half of your search. Since the key is chosen at random, there is no reason why it should necessarily be in the second half rather than the first. It's a coin toss: it's either in the first half or the second. So, you need to have half of the universe large enough

Caesar Cipher: Attacks

On average half of
all possible keys
must be tried to
achieve success

How is possible to **attack** the Caesar Cipher
without having the secret key)

- **Brute Force** attack: **we can try all the possible keys**
 - *is it feasible?* It depends on the **number of keys** (i.e. **key size**) and the **time required to check each key** (... computers are fast!)
- **Cryptanalytic** attack: (in this case) we can use **the frequency of characters** (or other **weaknesses** of think kind)
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Caesar Cipher: Attacks



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Caesar Cipher: Attacks



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How is possible to **attack** the C
without having the secret key)

This can be exploited
by the attacker!

How?

- **Brute Force** attack: we can
 - *is it feasible?* It depends on the time required to check each key (... computer power)
- **Cryptanalytic** attack: (in this case) we can use the frequency of characters (or other weaknesses of think kind)
 - in human languages **each letter** has a **different frequency** of use
 - the Caesar Cipher translates letters and therefore **it translates also their frequency**

In crittografia, questo significa che l'algoritmo non fornisce confusione: la struttura statistica del messaggio originale (il plaintext) rimane perfettamente visibile nel messaggio cifrato. DETTO NELLE SLIDES PRECEDENTI



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Symmetric Encryption

Symmetric Key Encryption



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Isn't obsolete.

- The universal technique for providing **confidentiality** for **transmitted** or **stored data**
- Two **requirements** for **secure use**:
 - a **strong encryption algorithm** against the three type of attacks that we saw
 - **sender** and **receiver** must have **obtained copies of the secret key**
in a secure fashion and **must keep the key secure**

Symmetric Key Enc

Use Cipher Algorithms that are standards

Please remember the **Kerckhoffs's principle**:

“A cryptosystem should be secure even if everything about the system, except the key, is public knowledge”

- The universal technique for secure use of symmetric key encryption is to store the key **separately from the stored data**
- Two **requirements** for **secure use**
 - a **strong encryption algorithm**
 - **sender** and **receiver** must have **obtained copies of the secret key in a secure fashion** and **must keep the key secure**

Symmetric Key Encryption



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- The universal technique for providing **confidentiality** for **transmitted** or **stored data**
- Two **requirements** for **secure use**:
 - a **strong encryption algorithm**
 - **sender** and **receiver** must have **obtained copies of the secret key**
in a secure fashion and **must keep the key secure**

Symmetric Key Encryption



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This is quite ridiculous: “to get a
secure communication channel
you need a secure channel”

to exchange the key to be used to encrypt/decrypt the message (to create a secure channel, i need a secure channel:
BIG LIMITATION, is not suited to be used in the internet because if the destination is far away it is difficult to find a
secure way to exchange key (Internet is insecure by nature, so attacker can get the key if they are listening))

confidentiality for transmitted or

- sender and receiver must have obtained copies of the secret key
in a secure fashion and must keep the key secure

Symmetric Key Encryption



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- The universal technique for **securely storing** **stored data**

- Two **requirements** for **secret communication**

- need a **strong encryption**

- **sender** and **receiver** must have **obtainable copies of the secret key**

in a secure fashion and **must keep the key secure**

We don't know if the key is leaked or not

The confidentiality of
communications depends on the
secrecy of the key

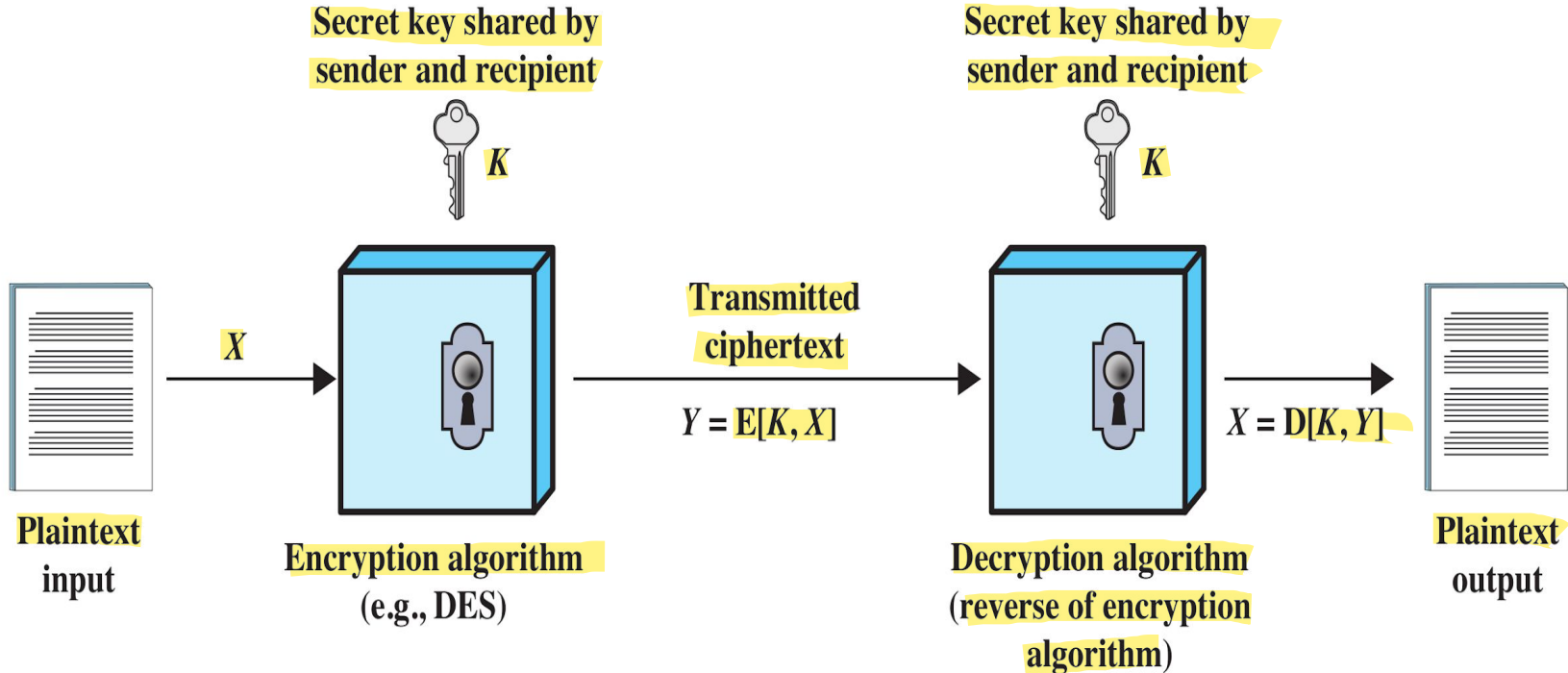
I have to trust even the destination because if it is malicious can give the key to others to encrypt/decrypt so there is a compromise of the communication channel

Simplified Model of Symmetric Encryption



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B
O
B



A
L
I
C
E

Simplified Model of Symmetric Encryption



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Secret message to be sent by

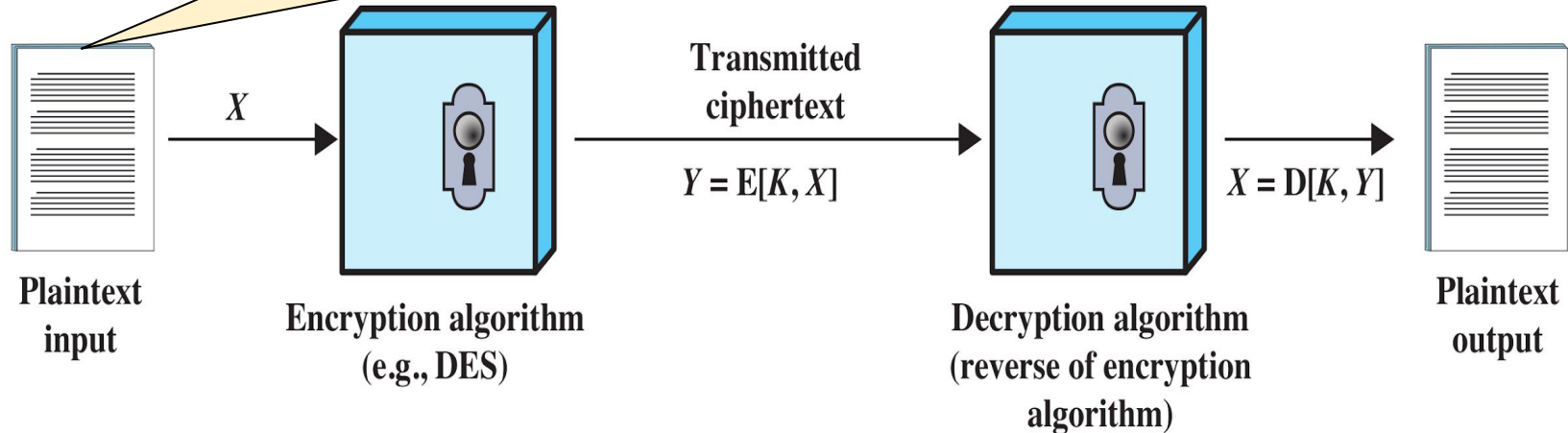
Bob to Alice. Bob wants

confidentiality

Secret key shared by
sender and recipient



A
L
I
C
E

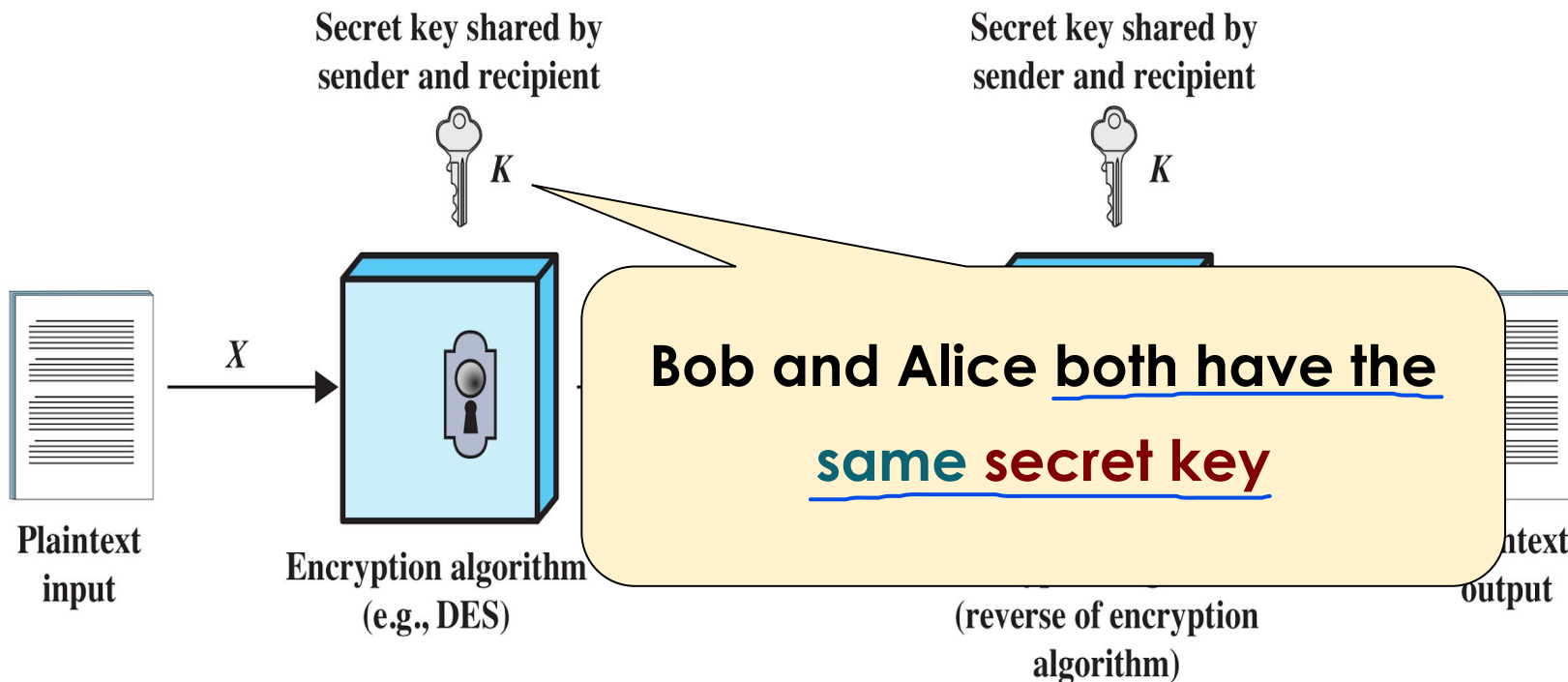


Simplified Model of Symmetric Encryption



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B
O
B



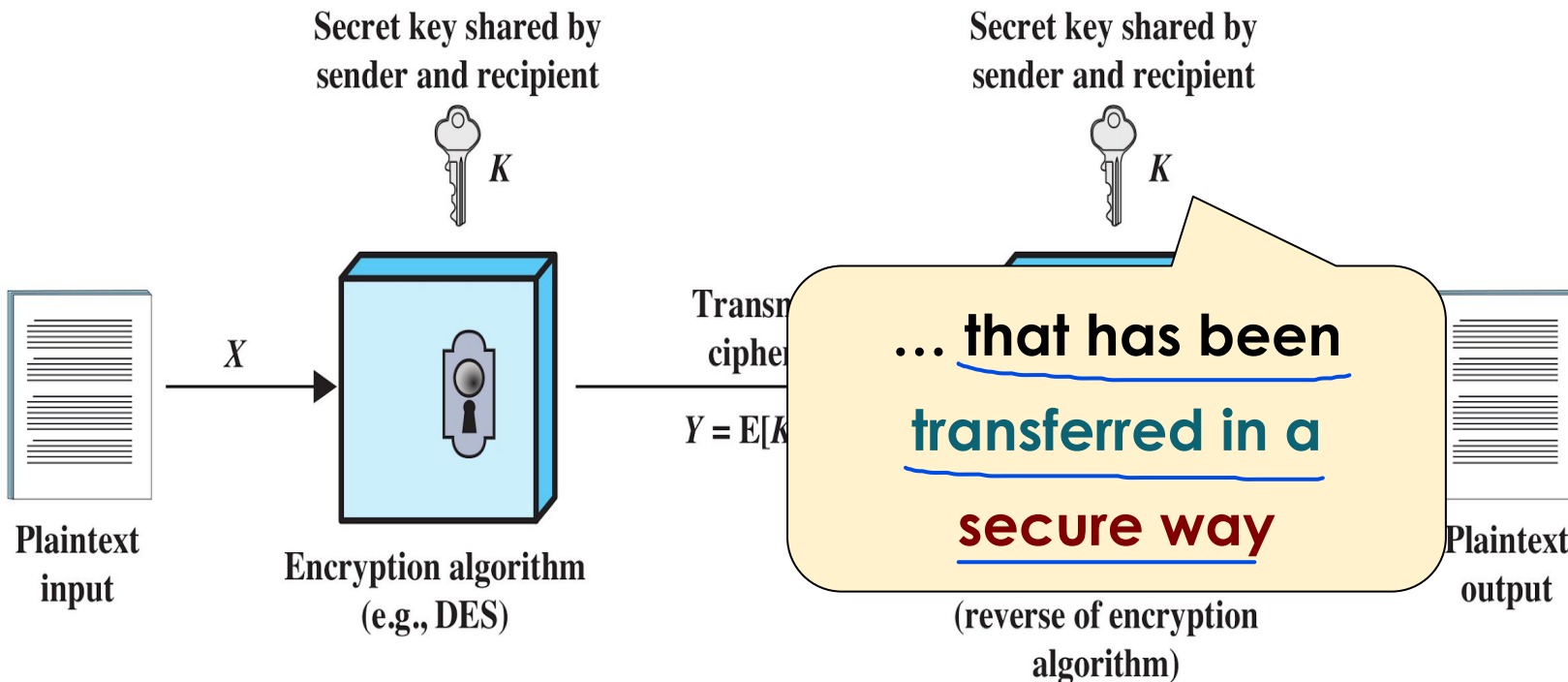
A
L
I
C
E

Simplified Model of Symmetric Encryption



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B
O
B



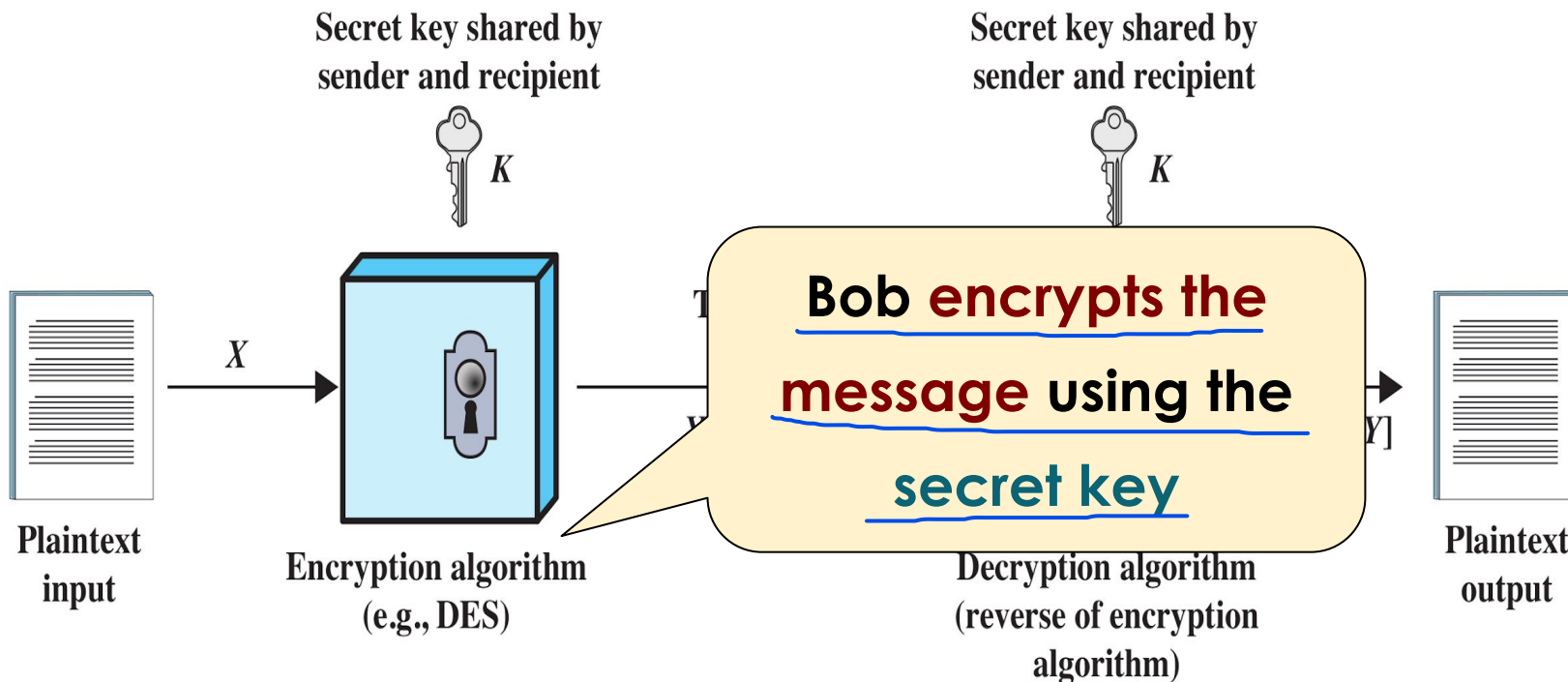
A
L
I
C
E

Simplified Model of Symmetric Encryption



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B
O
B



A
L
I
C
E

Simplified Model of Symmetric Encryption

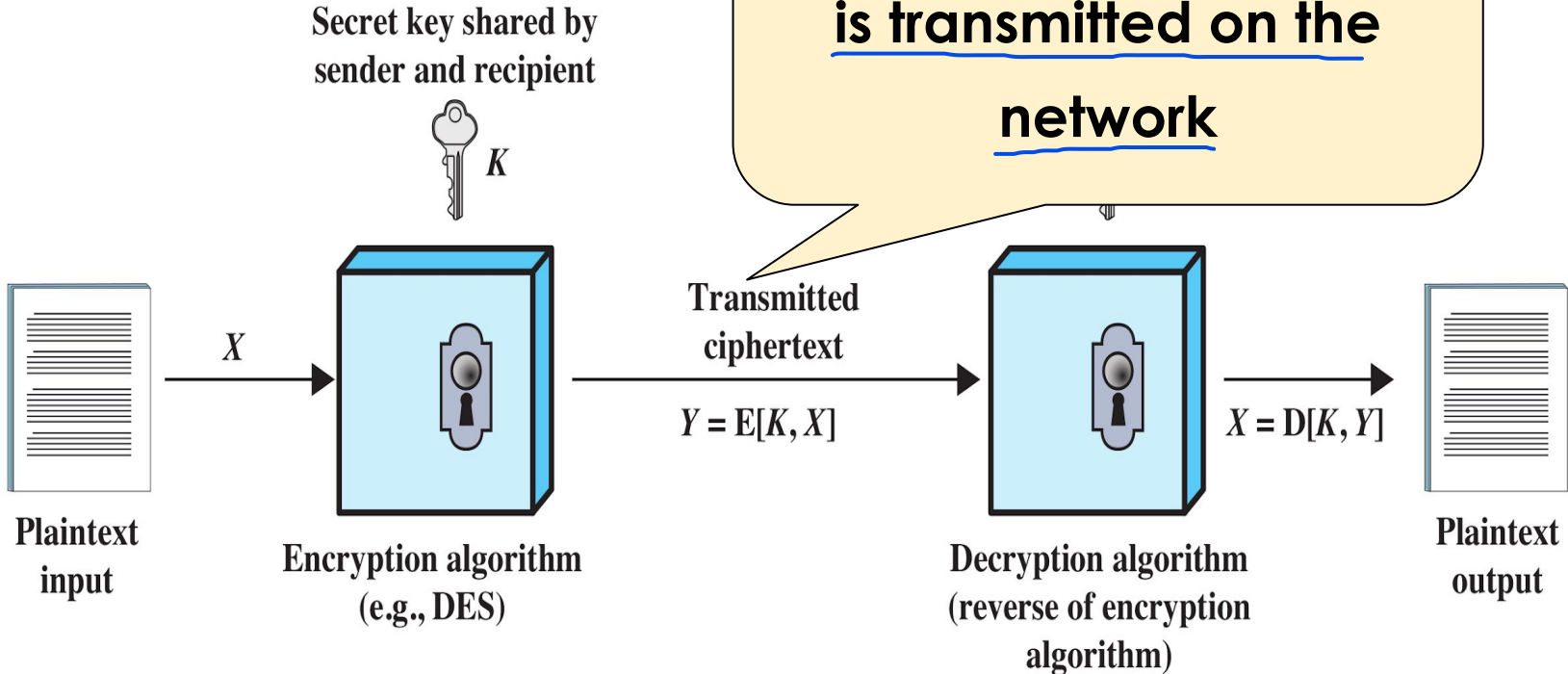


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The encrypted message
is transmitted on the
network

A
L
I
C
E

B
O
B



Simplified Model of Symmetric Encryption

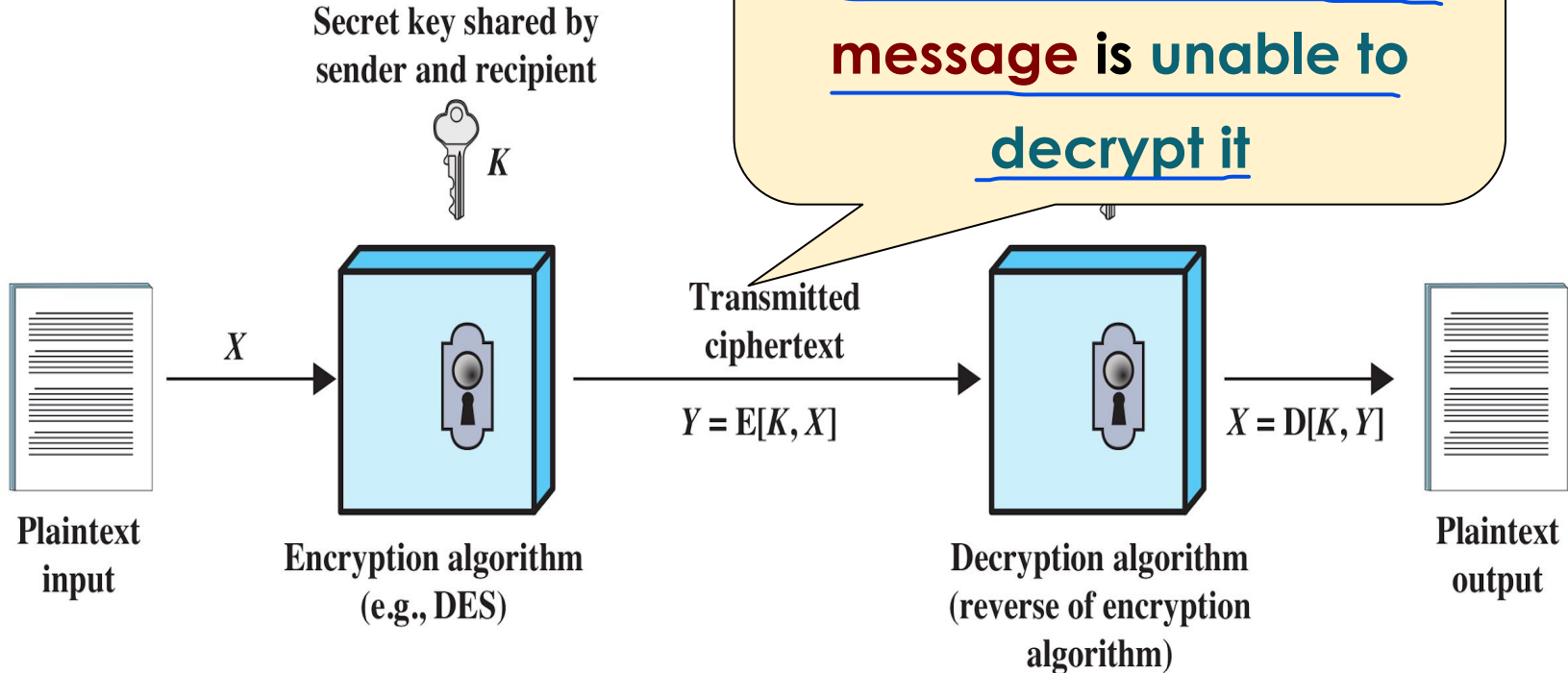


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A
L
I
C
E

An attacker that
intercepts the encrypted
message is unable to
decrypt it

B
O
B

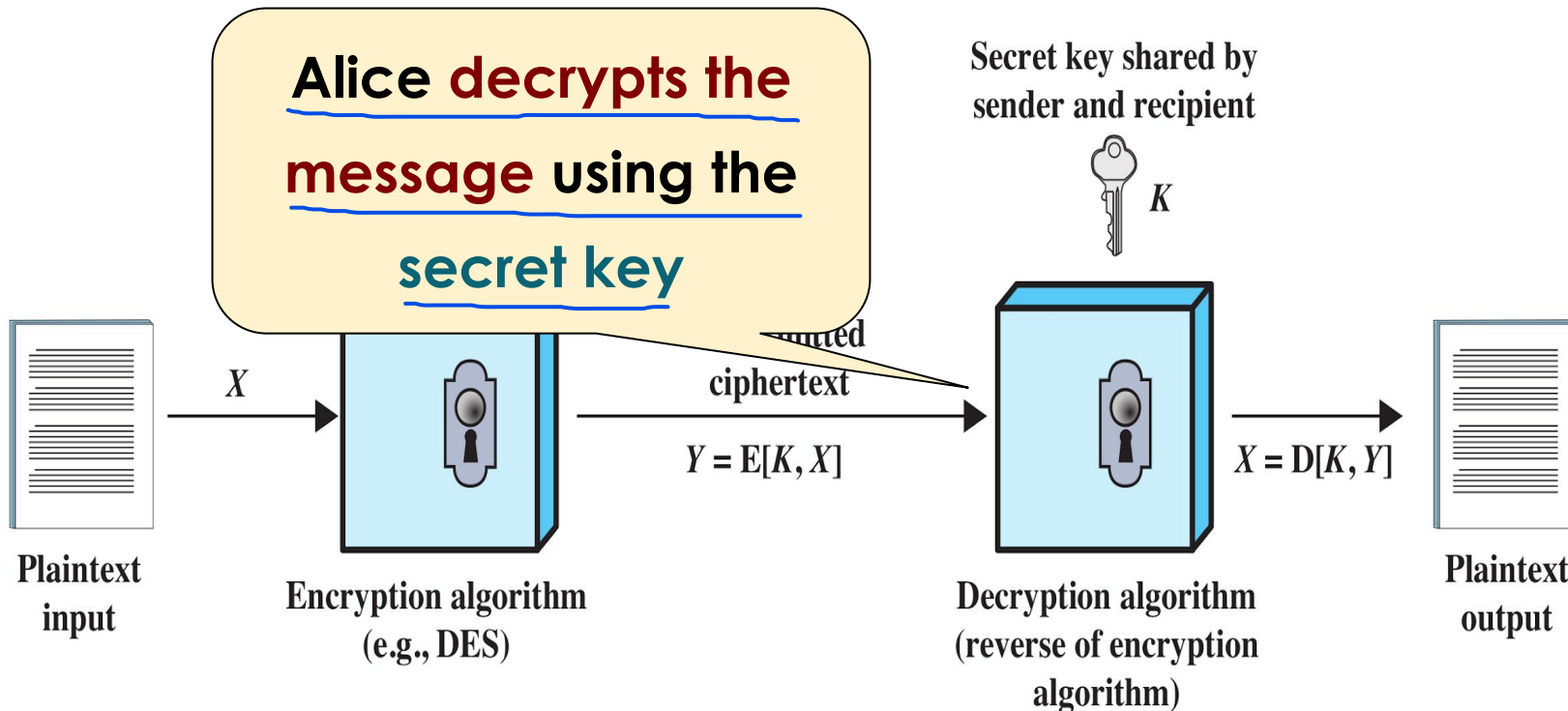


Simplified Model of Symmetric Encryption



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B
O
B



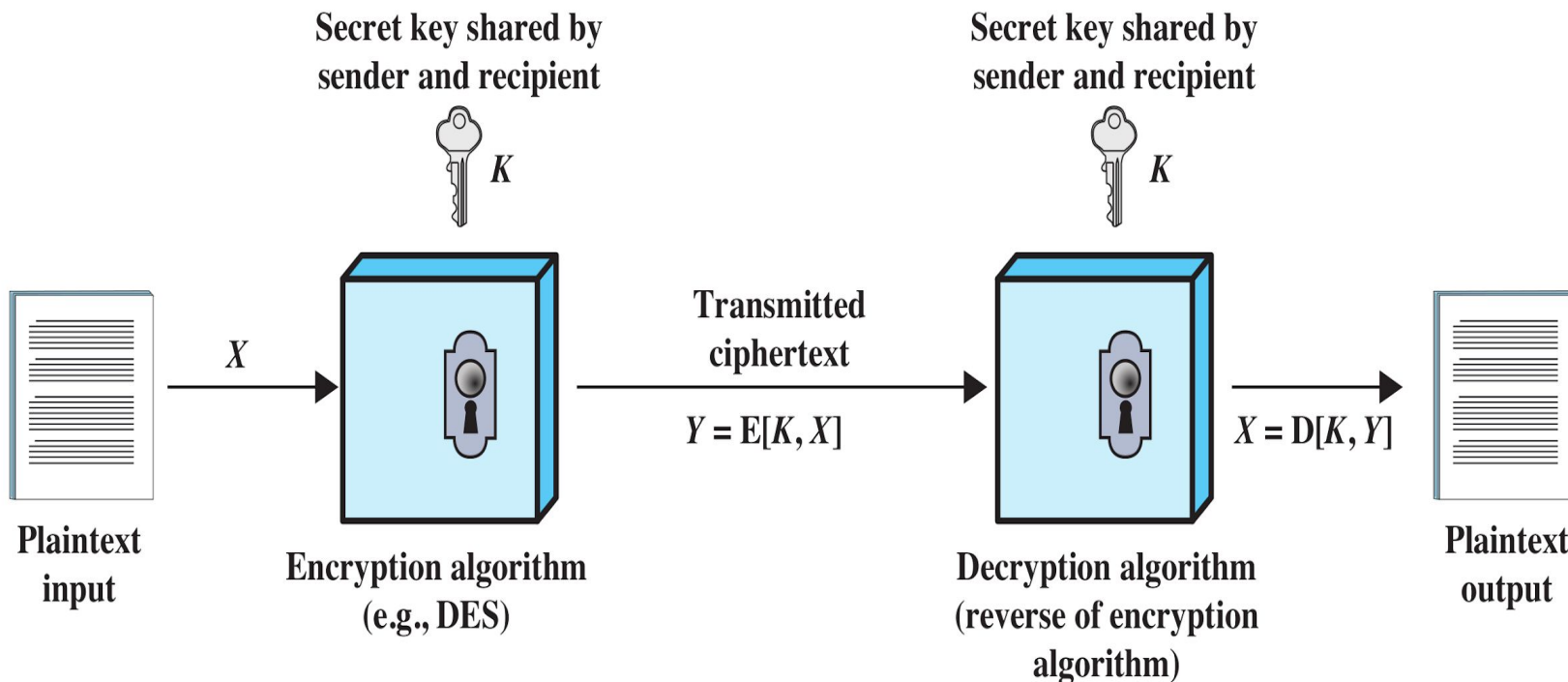
A
L
I
C
E

Simplified Model of Symmetric Encryption



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**B
O
B**



**A
L
I
C
E**

If a key is random, an attacker has zero information before they start. Every key in the universe (N) is equally likely. This forces the attacker into the "Worst Case" scenario: a slow, linear Brute Force attack. If the key isn't random (e.g., it's a word from a dictionary, a date of birth, or a common pattern), the attacker can use a Cryptanalytic Attack to bypass the "Universe of Keys."



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Symmetric Encryption: Attacks

The goal of Cipher Designers is to make this type of attack as costly as possible (if it is very costly, the attacker will not pursue this approach)

● **Brute Force** attack: **try all the possible keys**

The assumption that a key is random is fundamental because:

- It is the only way to guarantee that the attacker must perform a full Brute Force.
- It allows us to use Probability and Statistics to prove how long the data will stay safe.
- It prevents Dictionary Attacks and Pattern Recognition from providing a shortcut.

(fundamental assumption)

○ again, it is about the **key size** (if the key is **randomic** and **not a dictionary word**) and how fast is our computer (or cluster of computers)

● Attack using vulnerabilities caused by weak design of Ciphers

● **Cryptanalytic** attack: if the cipher is **weak** then it could be possible to find ways to decrypt the ciphertexts (i.e. WW2 Enigma Machine)

when build a Cipher we take into consideration computer speed of today and future computers to assess the strongness of a Cipher

● If the attack is **successful** then **all future and past messages** encrypted with that key are **compromised** (it is better to change frequently the key to avoid compromission (we don't know if the communication channel is compromised))



Before trying all the possible combinations of characters, I would try with all the words in dictionaries

- again, it is about the **key size** (if the key is **randomic** and **not a dictionary word**) and how fast is our computer (or cluster of computers)
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What is a

dictionary?

- If the cipher is **weak** then it could be possible to find ways to decrypt the ciphertexts (i.e. WW2 Enigma Machine)
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Before trying all the possible combinations of characters, I would try with all the words in dictionaries

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What is a dictionary?

- If the cipher is **weak** then it could be possible to find ways to decrypt the ciphertext (i.e. WW2 Enigma Machine)

What is the best dictionary?

- If the attacker has access to **future and past messages** encrypted with the same key...

Symmetric Encryption: Attacks



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- **Brute Force** attack: **try all the possible keys**
 - *again, it is about the **key size** (if the key is **randomic** and **not a dictionary word**) and how fast is our computer (or cluster of computers)*
- **Cryptanalytic** attack: if the cipher is **weak** then it could be possible to find ways to decrypt the ciphertexts (i.e. WW2 Enigma Machine)
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Symmetric Encryption: Attacks



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Key size (bits)	Number of alternative keys	Time required at 1 decryption/ μ s	Time required at 10^6 decryptions/ μ s
32	$2^{32} = 4.3 \times 10^9$	35.8 minutes	2.15 milliseconds
56	$2^{56} = 7.2 \times 10^{16}$	1142 years	10.01 hours
128	$2^{128} = 3.4 \times 10^{38}$	5.4×10^{24} years	5.4×10^{18} years
168	$2^{168} = 3.7 \times 10^{50}$	5.9×10^{36} years	5.9×10^{30} years
26 characters (permutation)	$26! = 4 \times 10^{26}$	6.4×10^{12} years	6.4×10^6 years

Average time required for **exhaustive key search** (i.e., Brute-Force)

Symmetric Encryption: algorithms



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- Many different **symmetric encryption algorithms** have been proposed
- It is better to use algorithms that are an **international standard**
- **DES** (Data Encryption Standard, 1977):
 - **key size** = 56 bits
 - It is **slow** and **very insecure** (the key size is too small)
- **3DES**: “execute 3 times the standard DES”
 - it is **very slow** but quite secure (key size = 56×3)
- **AES** (Advanced Encryption Standard, 2002):
 - **key size** = {128, 192 and 256} bits
 - very **fast** and **secure** (**now**... in the future?)

Symmetric Enc

Likely the most studied encryption algorithm: **some cryptanalytic issues but no relevant weaknesses**

- Many different
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In 1998, Electronic Frontier Foundation (EFF) announced that it had broken a DES in **< 3 days** with specialized hardware

Symmetric Enc



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56 bits (?!?)

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**It is not a power
of 2 :-)**

Symmetric Encryption: algorithms



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"**NSA** worked closely with **IBM** to strengthen the algorithm against all **except brute force attacks** and to strengthen substitution tables, called S-boxes. Conversely, NSA tried to convince IBM to **reduce the length of the key from 64 to 48 bits**. Ultimately they compromised on a **56-bit** key"

Thomas R. Johnson

Symmetric Encryption: algorithms



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 - it is **very slow** but quite secure (key size = 56×3)
- **AES** (Advanced Encryption Standard, 2002):
 - **key size** = {128, 192 and 256} bits
 - very **fast** and **secure** (**now**... in the future?)

Symmetric Encryption: algorithms



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- Many different **symmetric** algorithms are proposed
- It is better to use algorithms that are **fast** and **secure**
- **DES** (Data Encryption Standard, 1976):
 - **key size** = 56 bits
 - It is **slow** and **very insecure** (key size is too small)
- **3DES**: "execute 3 times the standard DES"
 - it is **very slow** but quite secure (key size = 56×3)
- **AES** (Advanced Encryption Standard, 2002):
 - **key size** = {128, 192 and 256} bits
 - very **fast** and **secure** (**now**... in the future?)

The **same hardware accelerators** used for DES can still be used with 3DES

Symmetric Encryption: algorithms



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- Many different **symmetric** algorithms have been proposed
- It is better to use algorithms with a larger block size
- **DES** (Data Encryption Standard, 1976)
 - **key size** = 56 bits
 - It is **slow** and **very insecure** (the **block size** is too small)
- **3DES**: "execute 3 times the standard DES"
 - it is **very slow** but quite secure (key size = 56×3)
- **AES** (Advanced Encryption Standard, 2002):
 - **key size** = {128, 192 and 256} bits
 - very **fast** and **secure** (**now**... in the future?)

The **block size** is **too small**
(see next slides)

Symmetric Encryption: algorithms



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	DES	Triple DES	AES
Plaintext block size (bits)	64	64	128
Ciphertext block size (bits)	64	64	128
Key size (bits)	56	112 or 168	128, 192, or 256

Symmetric Encryption: **block ciphers**



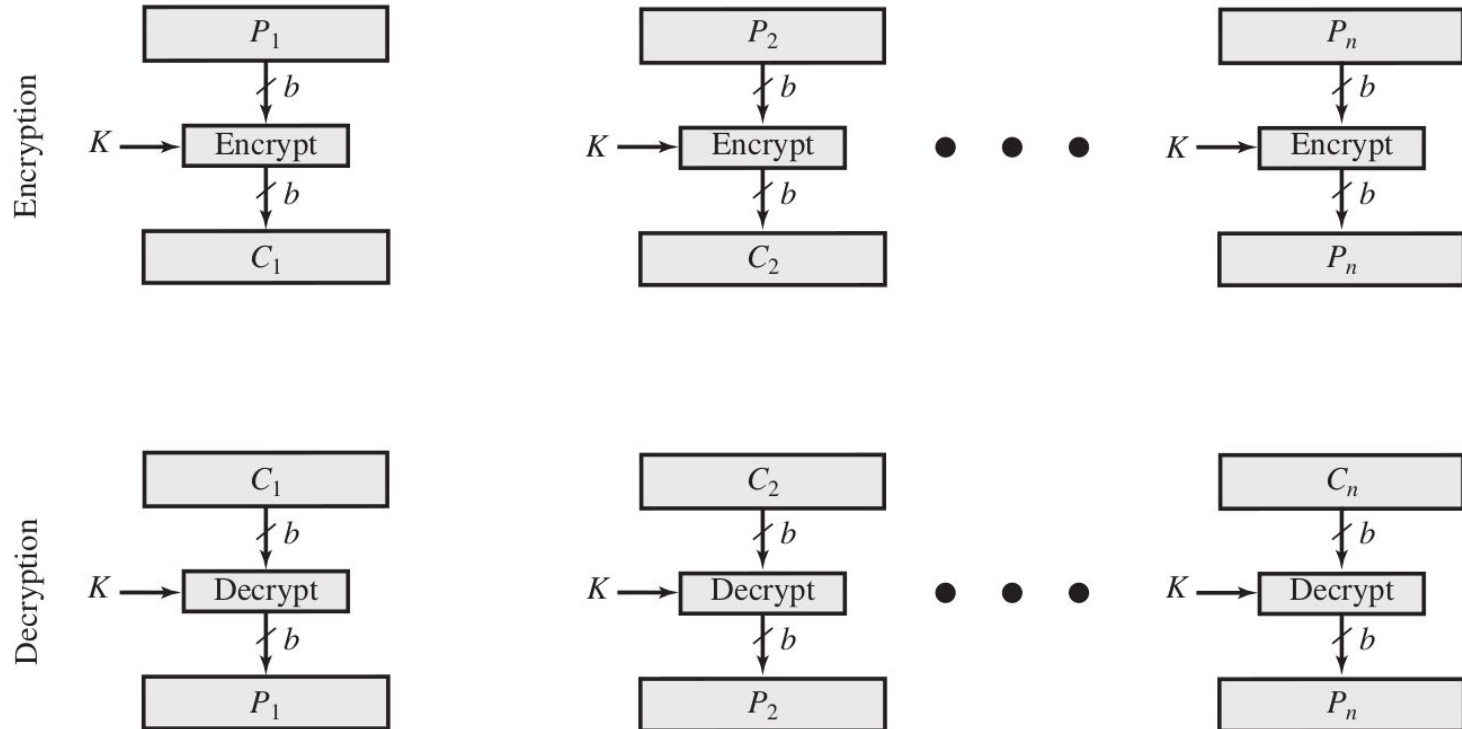
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- **DES**, **3DES** and **AES** are **symmetric block ciphers**
- Typically data units (e.g., messages or files) are **larger than a single 64-bit or 128-bit block**
- Possible solution... **Electronic CodeBook (ECB)** mode:
 - (the simplest approach to multiple-block encryption)
 - **divide** the input data in a **sequence of blocks**
 - apply **padding** to the last block (if necessary)
 - each block is **encrypted** using **the same key**
 - (**regularities** in the plaintext could be exploited for **cryptanalysis**)

Modes of Operation: ECB



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Modes of Operation: ECB



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- Consider the **advantages** vs. **disadvantages** of **ECB** in terms of:
 - **performance** (e.g., encryption throughput)
 - potential **attacks** (e.g., message **reordering** and **replay** attacks)
 - there is a huge problem of “**diffusion**” (see the next slide)
- Other “more complex” **modes of operation** are available (**at a cost**)

Modes of Operation: ECB



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Original image



Encrypted using ECB mode

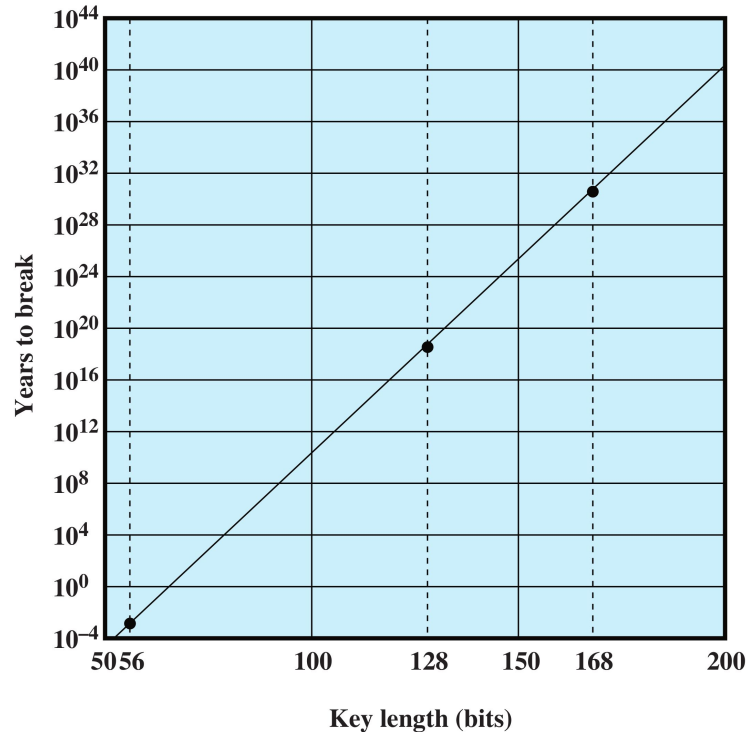


Modes other than ECB result in
pseudo-randomness



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Brute-Force



Time To Break a Code

(assuming 10^6 decryptions/ μ s)

Is this assumption realistic?

Check it out!

T2

Symmetric Encryption: **block ciphers**



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- **DES**, **3DES** and **AES** are symmetric **block** ciphers

How can we deal with data that is
NOT produced in blocks?



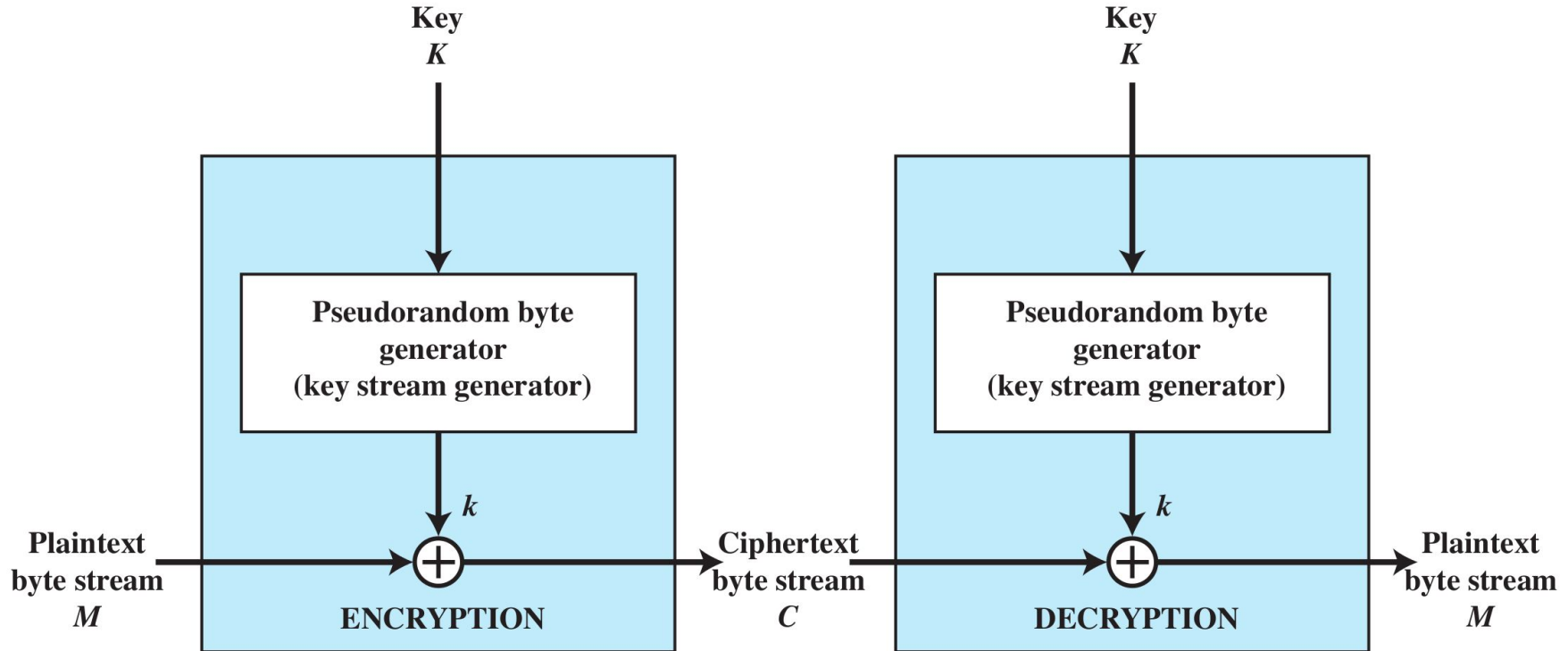
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Stream Encryption

Stream Encryption

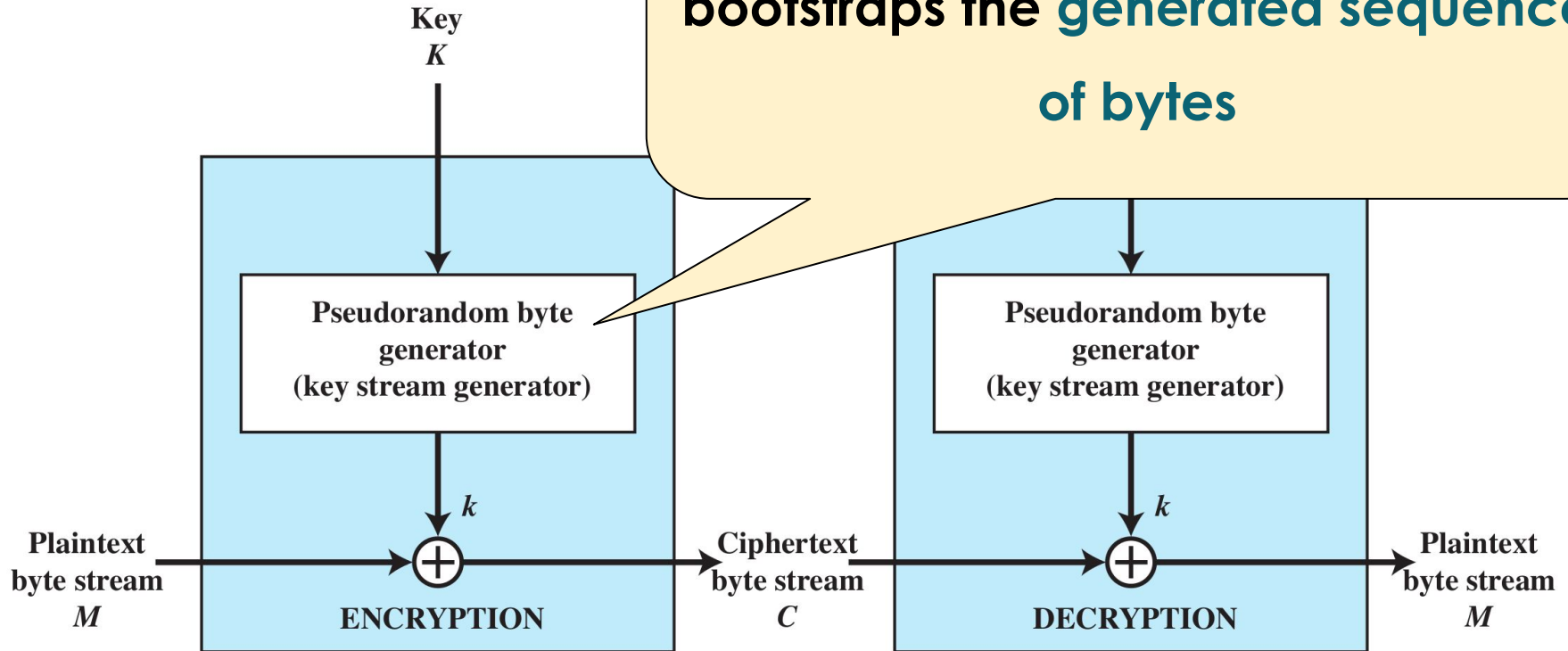


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Stream Encryption

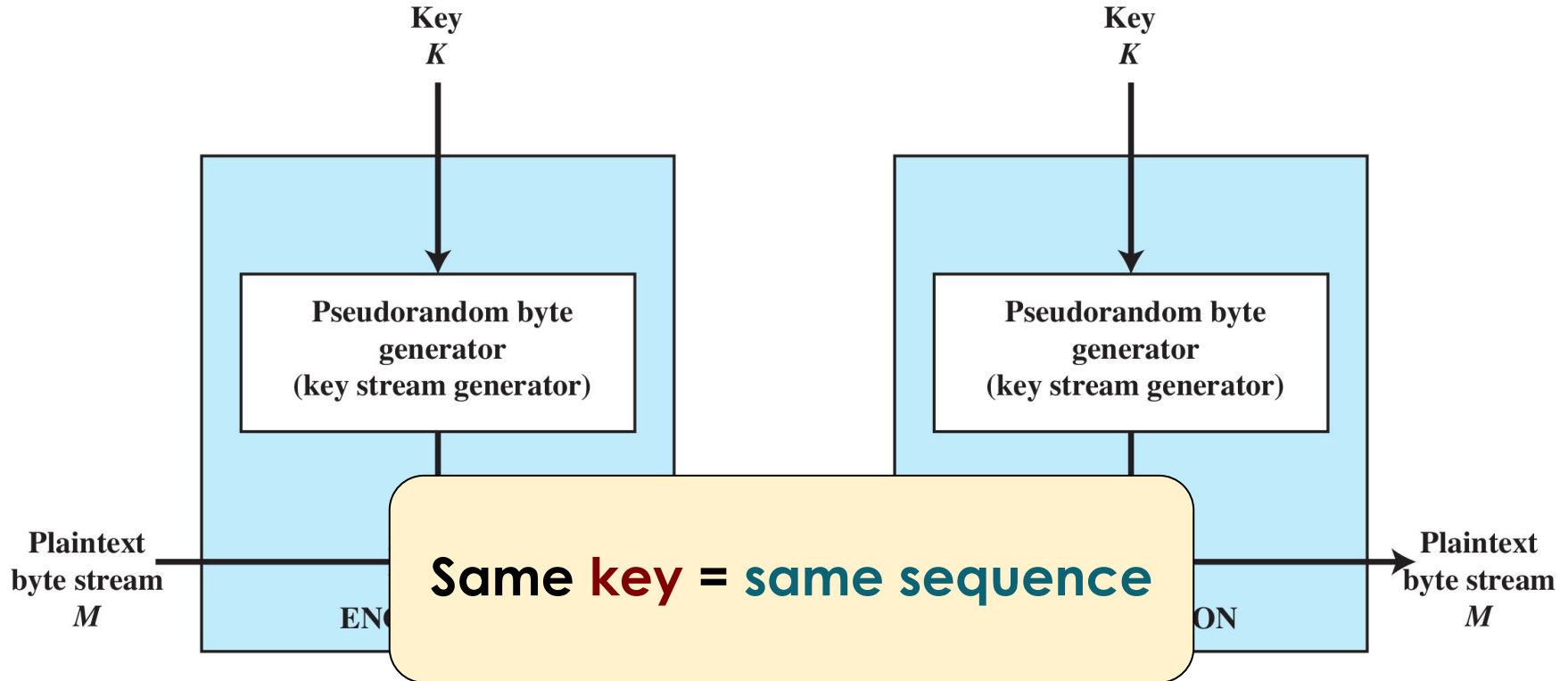
The **key** is the parameter that bootstraps the **generated sequence of bytes**



Stream Encryption



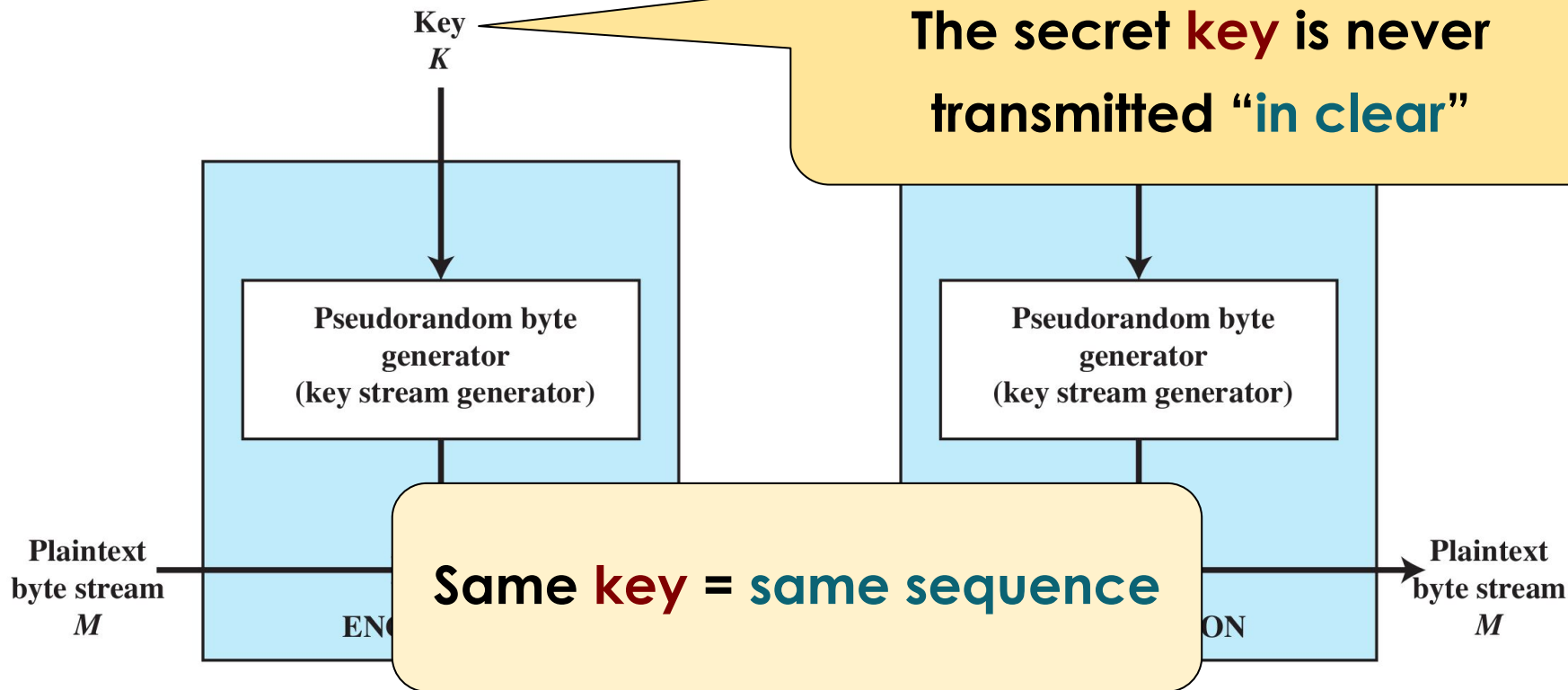
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Stream Encryption



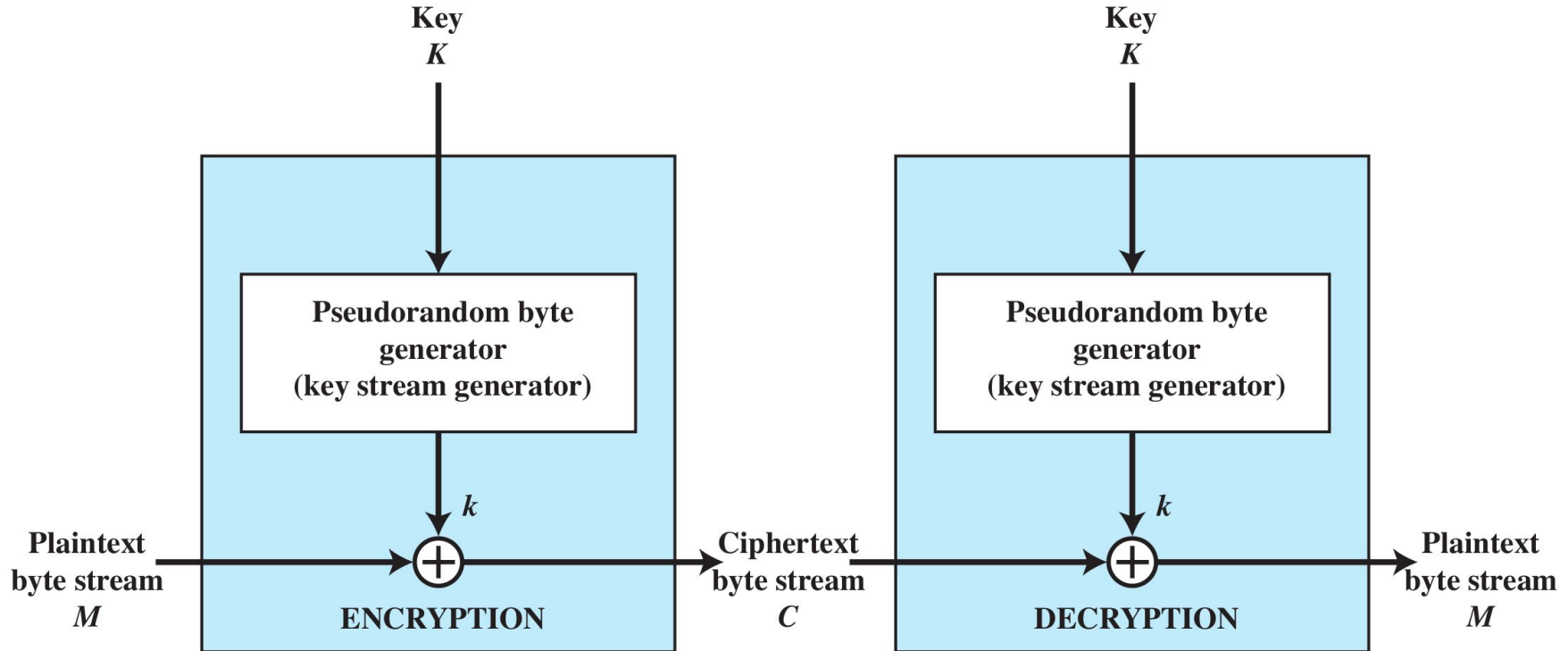
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Stream Encryption



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Block **vs.** Stream Ciphers

Block **vs.** Stream Ciphers



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- **Block** ciphers:
 - process the **input** one block of elements at a time
 - produce an **output** block for each input block
 - can **reuse keys** (and usually they do that)
 - can require to add **padding**
 - bad for **efficiency**!
 - how is **padding** generated?
 - are **more common** (than stream ciphers)

Block **vs.** Stream Ciphers



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- **Stream** ciphers:
 - process the **input** elements **continuously**
 - produces **output one element at a time**
 - almost always **faster** and **use far less code**
 - encrypts plaintext **one byte at a time**
 - pseudorandom stream is one that is **unpredictable without knowledge of the input key**



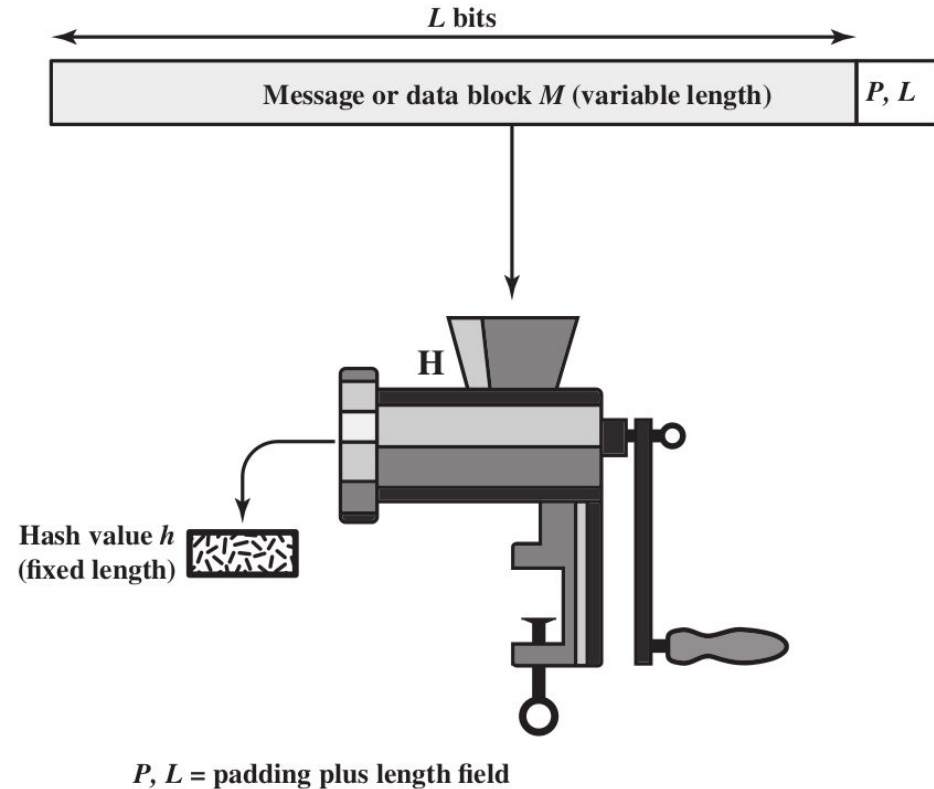
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Secure Hash Functions

Secure Hash Functions



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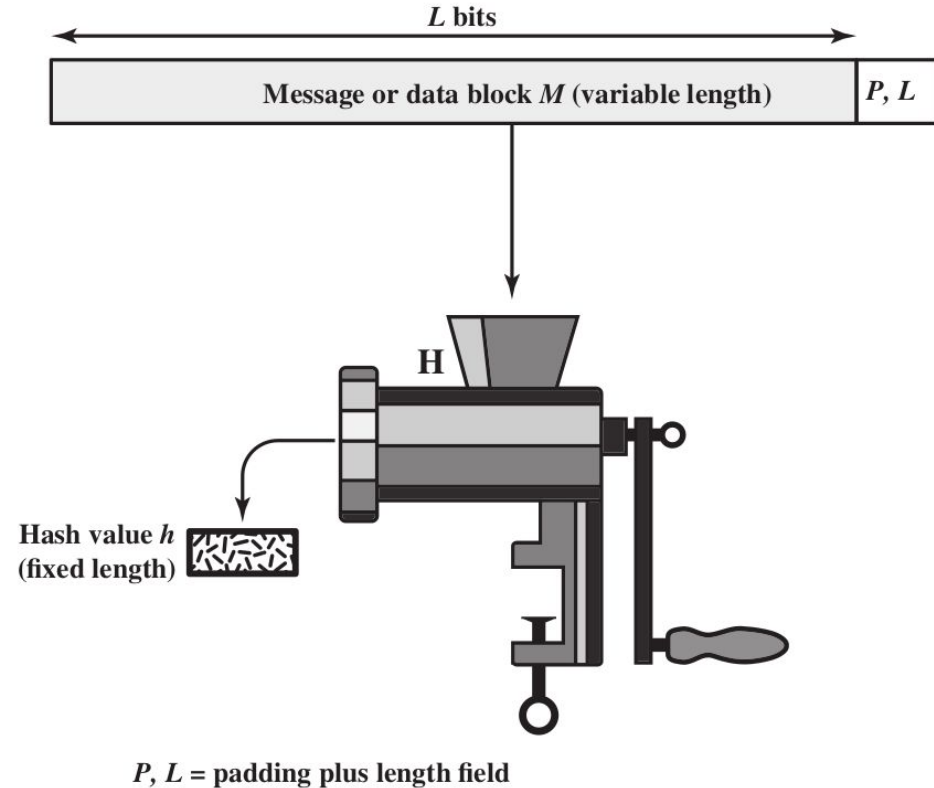


Secure Hash Functions



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The **input size** is **variable**



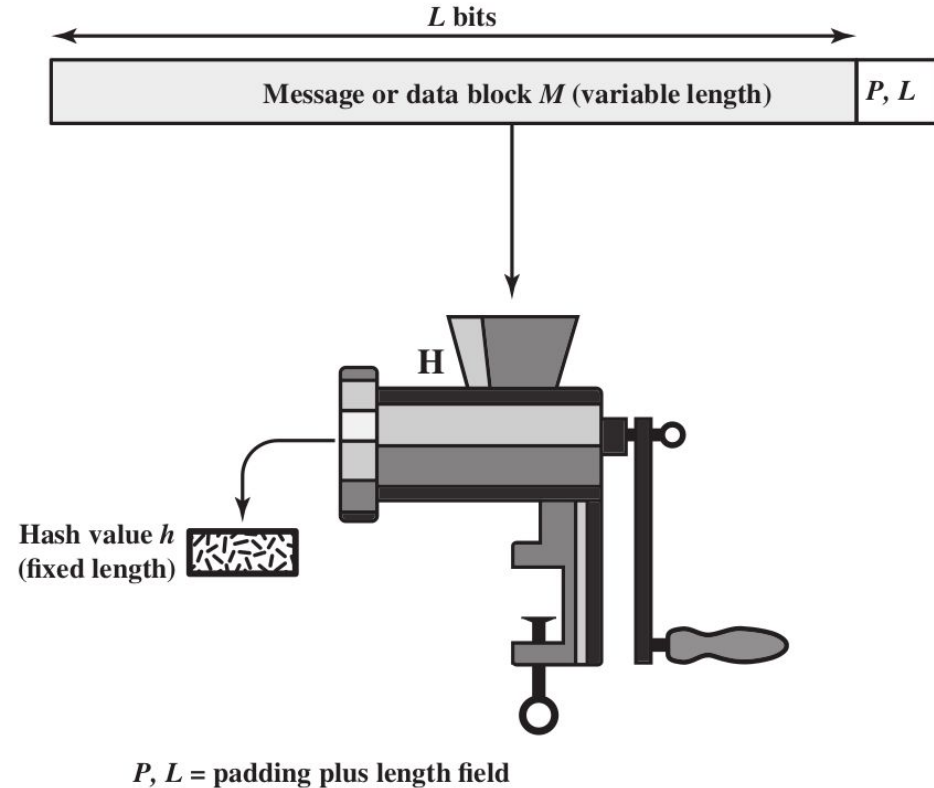
Secure Hash Functions



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The **input size** is **variable**

H is a mathematical function



Secure Hash Functions

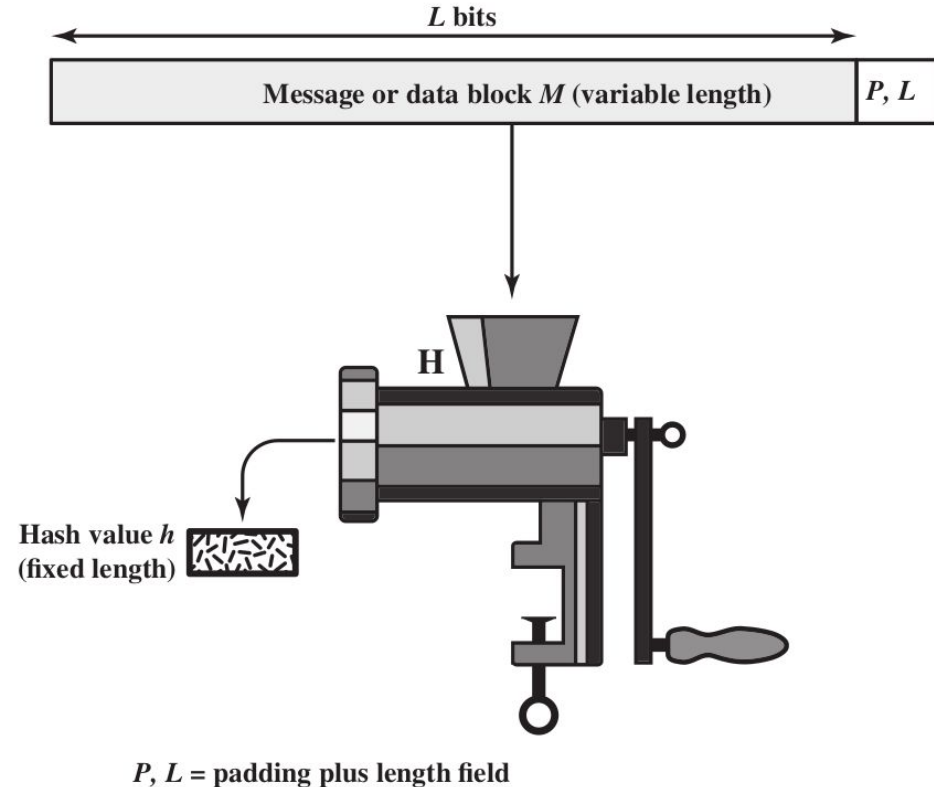


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The **input size** is **variable**

H is a mathematical function

The **output size** is **fixed**



Secure Hash Functions: **properties**



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1. H can be applied to a block of data of any size
2. H produces a fixed-length output
3. $H(x)$ is **relatively easy to compute** for any given x
4. For any given code h , it is computationally infeasible to find x such that $H(x) = h$ (**one-way** or **preimage resistant**)
5. For any given block x , it is computationally infeasible to find $y \neq x$ with $H(y) = H(x)$ (**second preimage resistant** or **weak collision resistant**)
6. It is computationally infeasible to find **any pair** (x, y) such that $H(x) = H(y)$ (**strong collision resistant**)

Secure Hash Functions: **security**



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- Approaches to **attack** a **secure hash function**
 - **cryptanalysis**: exploits **logical weaknesses** in the algorithm
 - **brute-force**: the strength of the hash function depends solely on the **length of the hash code** produced by the algorithm
- **SHA** (Secure Hash Algorithm) is the most widely used hash function
 - many different “families” (SHA0, SHA1, SHA2, SHA3) and hash value lengths

Secure Hash Functions: **security**



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- Approaches to **attack** a **secure hash function**

- **crypt**

- **brut**

the

- **SHA** (Se

- many different “families” (SHA0, SHA1, SHA2, SHA3) and hash value lengths

Lifetime of some famous
cryptographic hash functions

Secure Hash Functions: applications



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- **Passwords:**

- the operating systems (and most online services) **store only** the
hash of passwords

- **Intrusion detection:**

- somehow **compute** and **securely store** the **hash code of relevant files** in the system. This permits to verify their **integrity**

Secure Hash Functions: applications



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- **Passwords:**

- the operating systems (and most online services) **store only** the
hash of passwords

- **Intrusion detection:**

- somehow **compute** a
files in the system. This

Plaintext passwords are never stored. Why? To be discussed in the authentication chapter

Secure Hash Functions: applications



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- **Passwords:**
 - the operating systems (and most online services) **store only** the **hash of passwords**
- **Intrusion detection:**
 - somehow **compute** and **securely store** the **hash code of relevant files** in the system. This permits to verify their **integrity**

Secure Hash Functions: applications



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- **Passwords:**

- the operating system (and most online services) store **only the hash of password**

This operation is quite costly
in terms of **computation**

- **Intrusion detection**

- somehow **compute** and **securely store** the **hash code of relevant files** in the system. This permits to verify their **integrity**

Secure Hash Functions: applications



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- **Passwords:**

- the operating system

hash of passwords

It is **not so easy!** Why?

For different reasons...

- **Intrusion detection:**

- somehow **compute** and **securely store** the **hash code of relevant**

files in the system. This permits to verify their **integrity**

Secure Hash Functions: applications



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- **Passwords:**

- the operating systems (and applications) store the

hash of passwords

**Why do we not check all
the files in the system?**

- **Intrusion detection:**

- somehow **compute** and **securely store** the **hash code of relevant**

files in the system. This permits to verify their **integrity**

Secure Hash Functions: applications



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- **Passwords:**

- the operating systems

hash of passwords

It works... if the **tool used for verifying the integrity is not compromised**

- **Intrusion detection:**

- somehow **compute** and **securely** the **hash code of relevant**

files in the system. This permits to verify their **integrity**



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Dictionary Attack

DICTIONARY ATTACK!



Practical **dictionary attack** to
the symmetric encryption

How does it works?

Check it out!

T3

Symmetric Key Encryption



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This is quite ridiculous: “to get a *secure communication channel* you need a *secure channel*”

- sender and receiver must have obtained

in a secure fashion and must keep the key secure

confid

How is possible to implement *secure communication* in online banking or e-commerce with such a limitation?



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Public-Key Encryption

Public-Key Encryption



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- **Bits of history**

- **1969** - United Kingdom - James Ellis (GCHQ): **theoretical basis**
- **1976** - USA - Whitfield Diffie and Martin Hellman: **Diffie–Hellman key exchange** (solution to the key distribution problem)
- **1977** - USA - **R**onald Rivest, Adi **S**hamir and Leonard **A**delman: **RSA public-key cryptosystem**
- **1991** - USA - Phil Zimmermann, **Pretty Good Privacy (PGP)**

Public-Key Encryption



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- **Bits of history**

- **1969** - United Kingdom - James Ellis (GCHQ): **theoretical basis**
- **1976** - USA - Whitfield Diffie and Martin Hellman: **key exchange** (solution to the key distribution problem)
- **1977** - USA - Ronald Rivest, Adi Shamir, and Leonard Adleman: **public-key cryptosystem**
- **1991** - USA - Phil Zimmermann, **Pretty Good Privacy (PGP)**

**Classified (i.e., secret) and
disclosed only in 1997**

SA

Public-Key Encryption



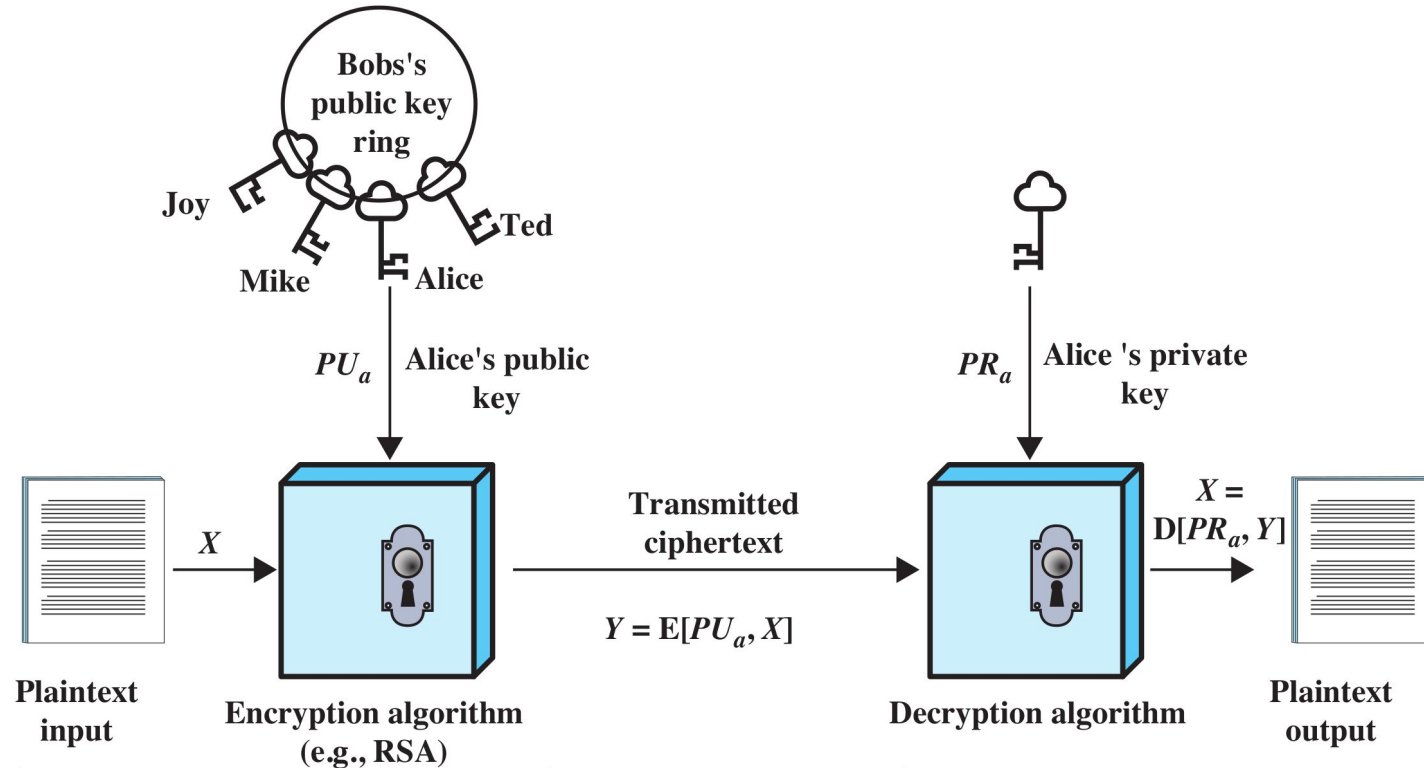
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- The **public-key encryption** is based on some **complex mathematical problems**
- It is a form of **asymmetric encryption** since **every user has 2 keys**:
 - one **public key**: *that is made as public as possible*
 - one **private key**: *that is maintained carefully secret*
- Every **couple of keys** is **generated together**, the two keys are **linked by some mathematical properties**

Public-Key Encryption



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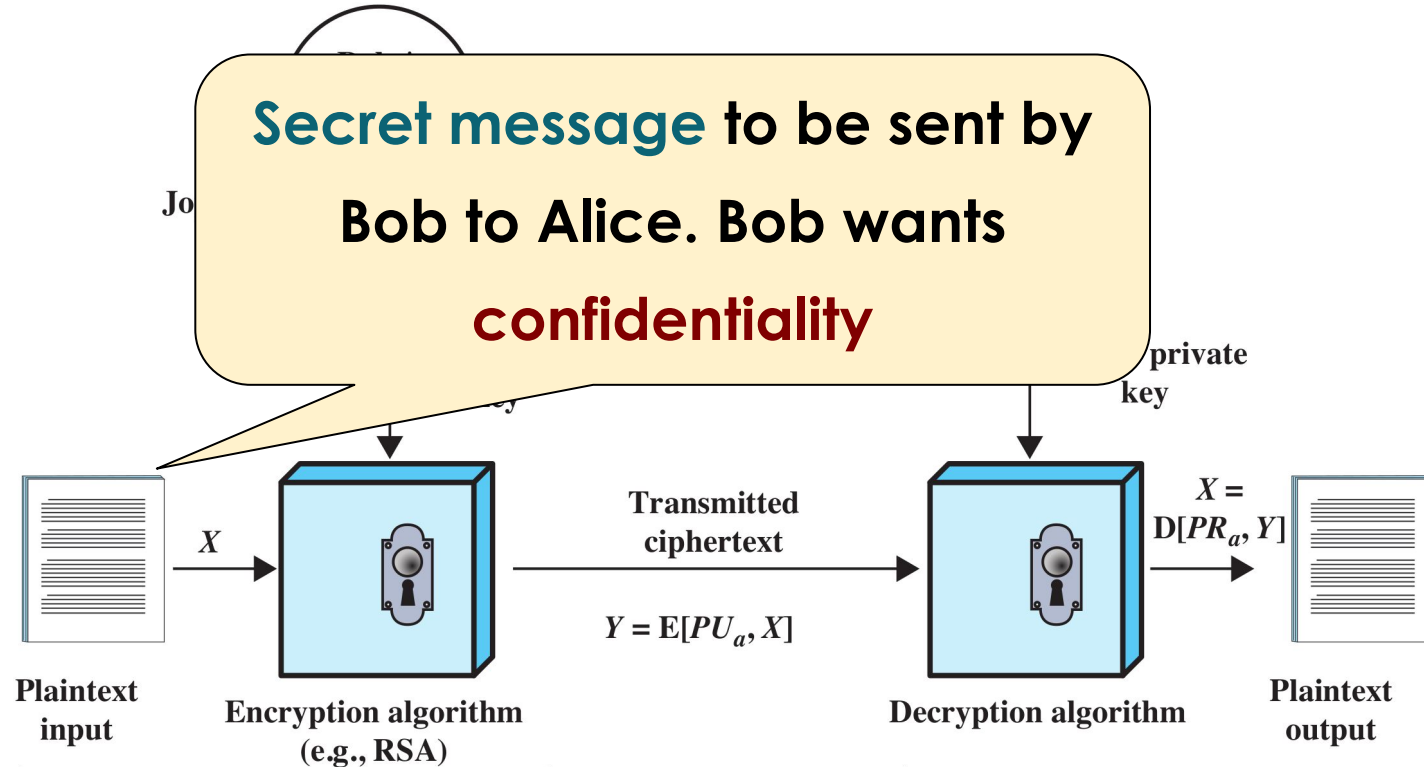
B
O
B

A
L
I
C
E

Public-Key Encryption



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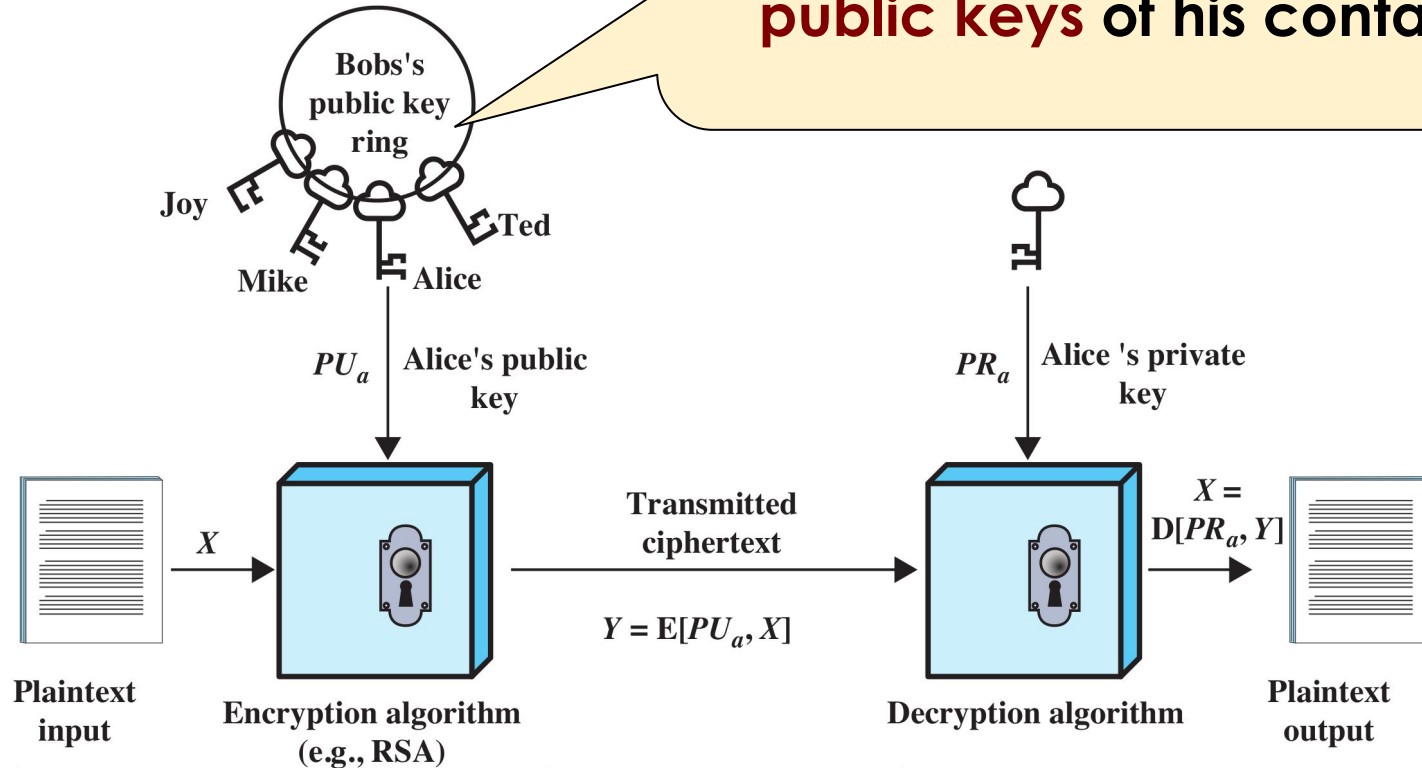


B
O
B

A
L
I
C
E

Public-Key Encryption

Bob has a **keyring** with all the **public keys** of his contacts



B
O
B

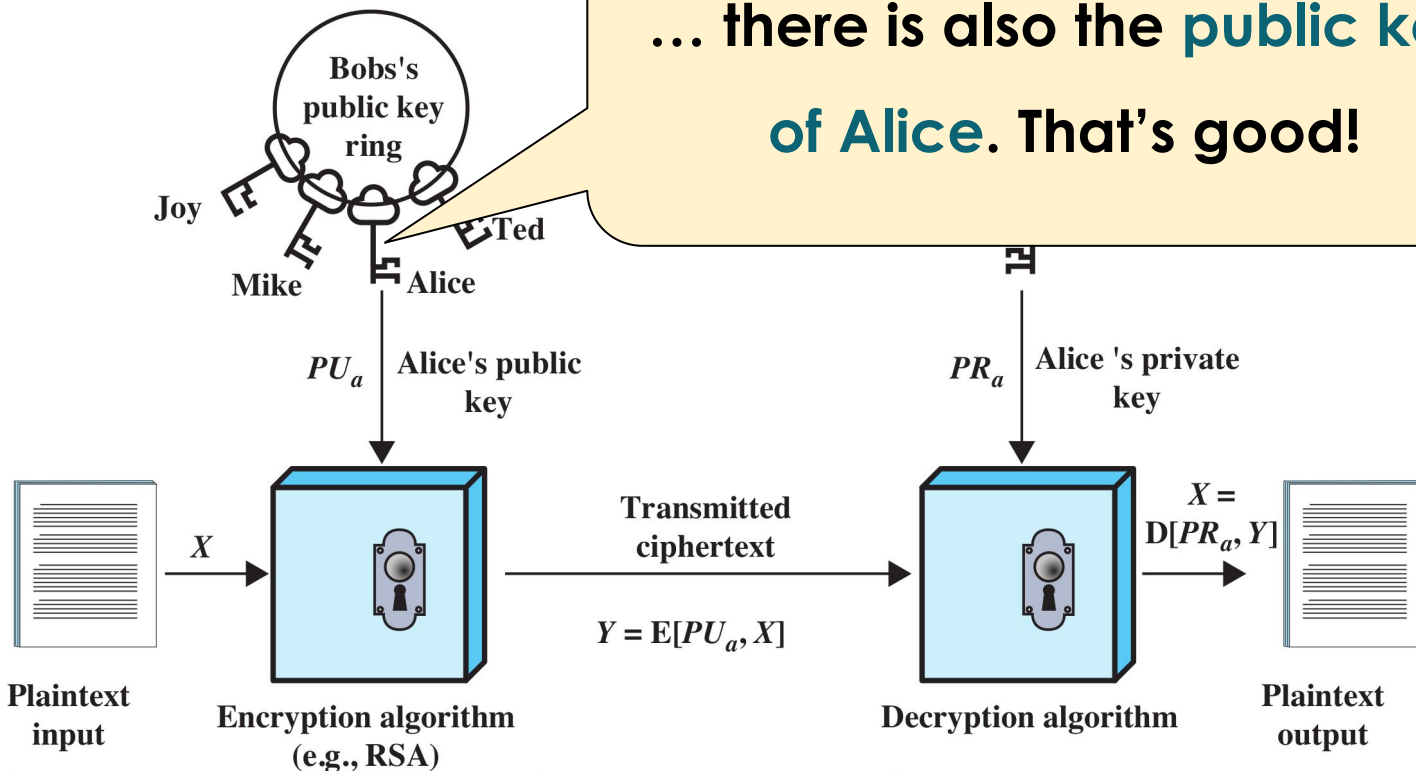
A
L
I
C
E

Public-Key Encryption



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... there is also the **public key**
of Alice. That's good!



B
O
B

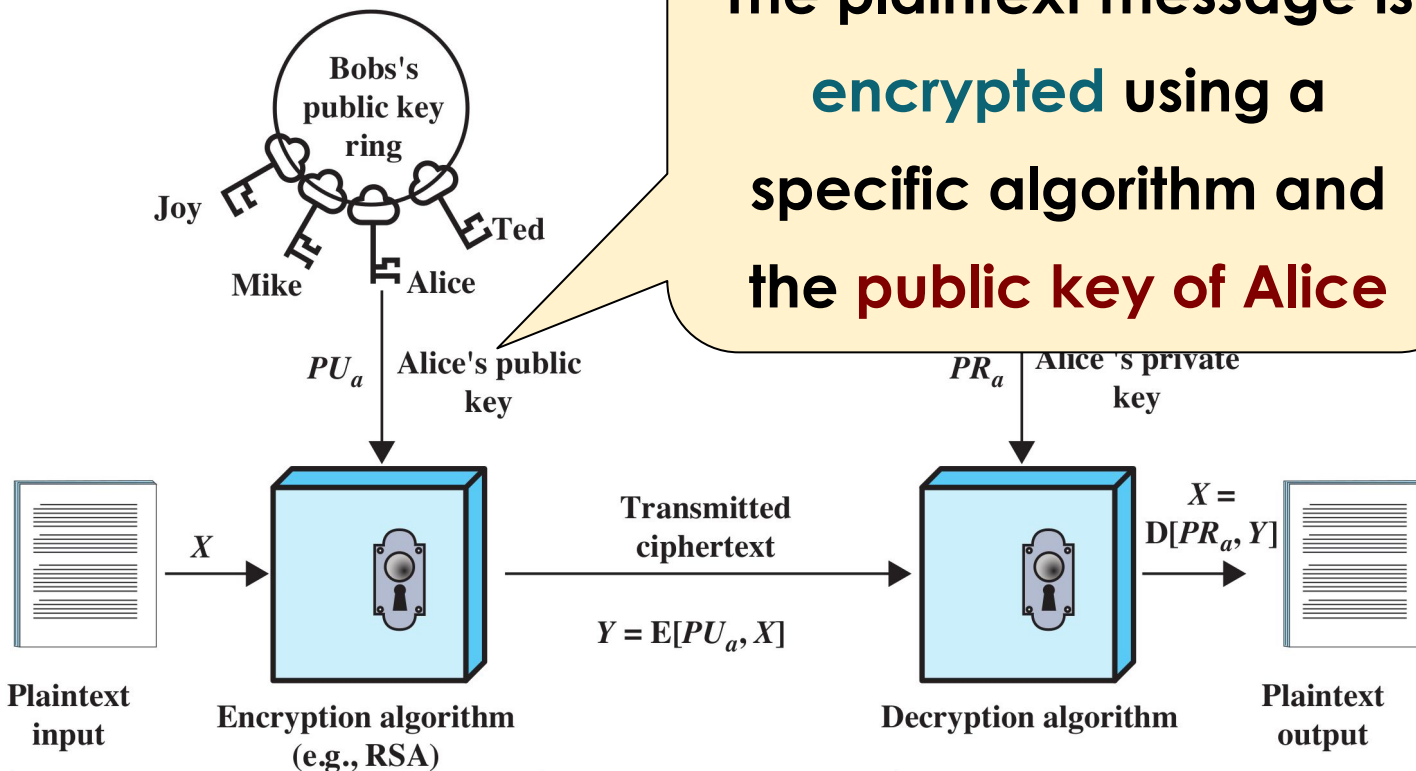
A
L
I
C
E

Public-Key Encryption



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The plaintext message is
encrypted using a
specific algorithm and
the **public key of Alice**

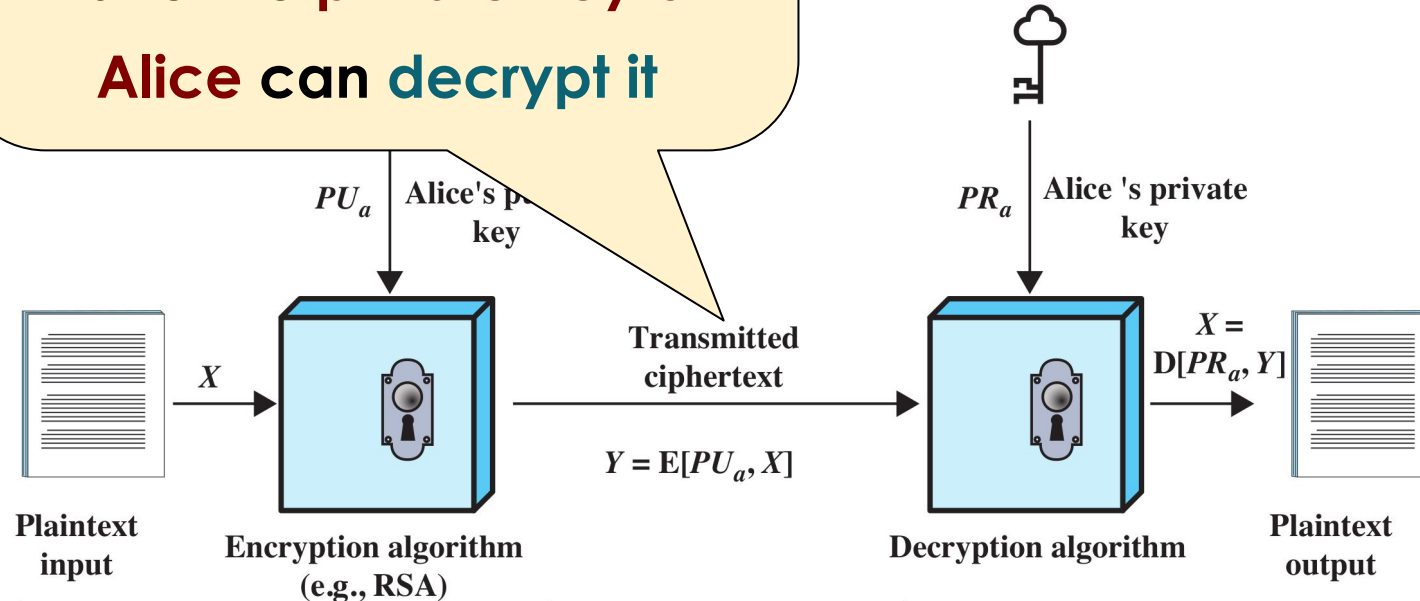


B
O
B

A
L
I
C
E



The ciphertext can be transmitted on the Internet since only those who have the **private key of Alice** can **decrypt it**



A
L
I
C
E

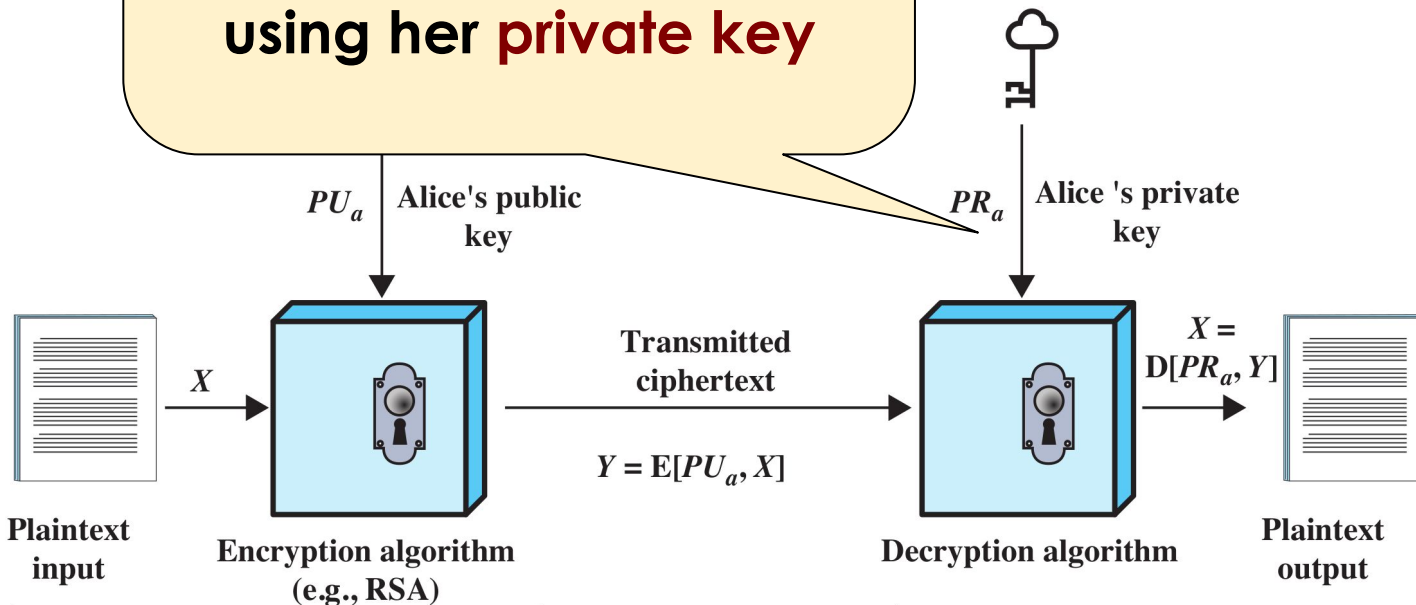
B
O
B

Public-Key



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Alice decrypts the
encrypted message
using her **private key**



B
O
B

A
L
I
C
E

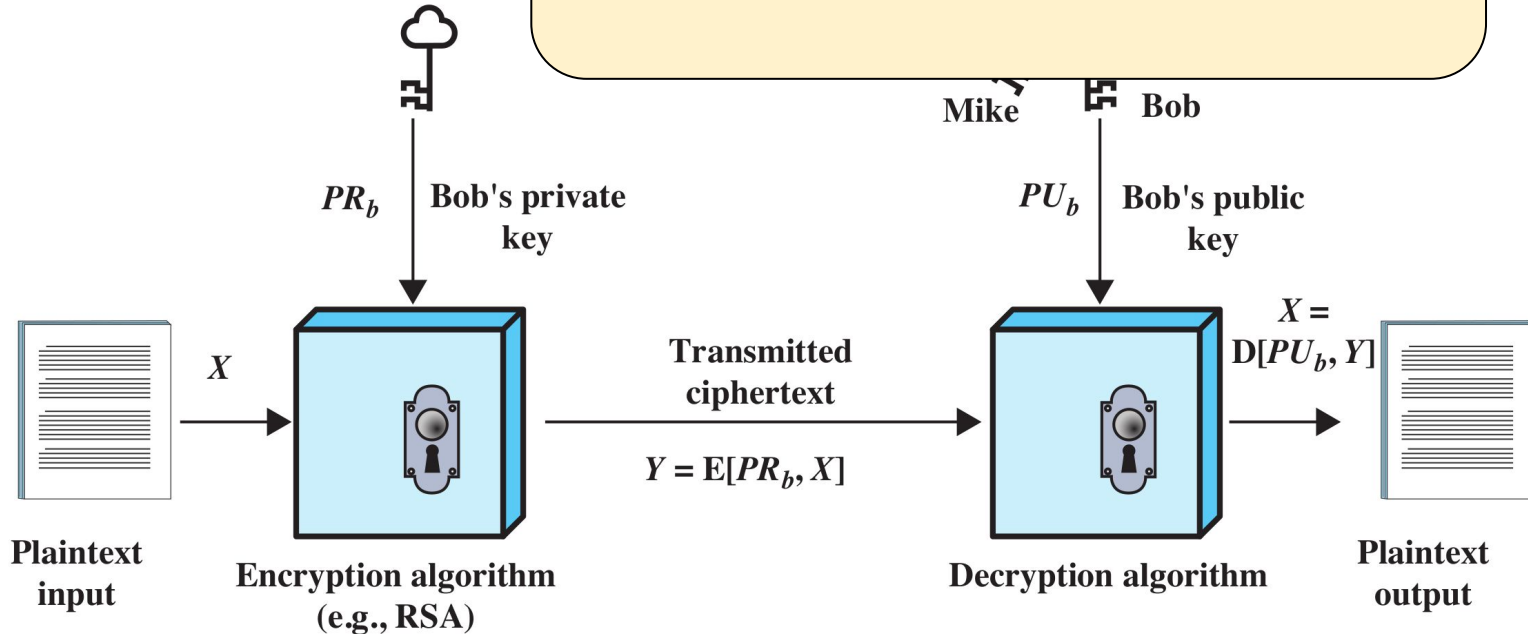
Public-Key Encryption



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Public key encryption can be
also used **in a different way**

A
L
I
C
E



B
O
B

Public-Key Encryption

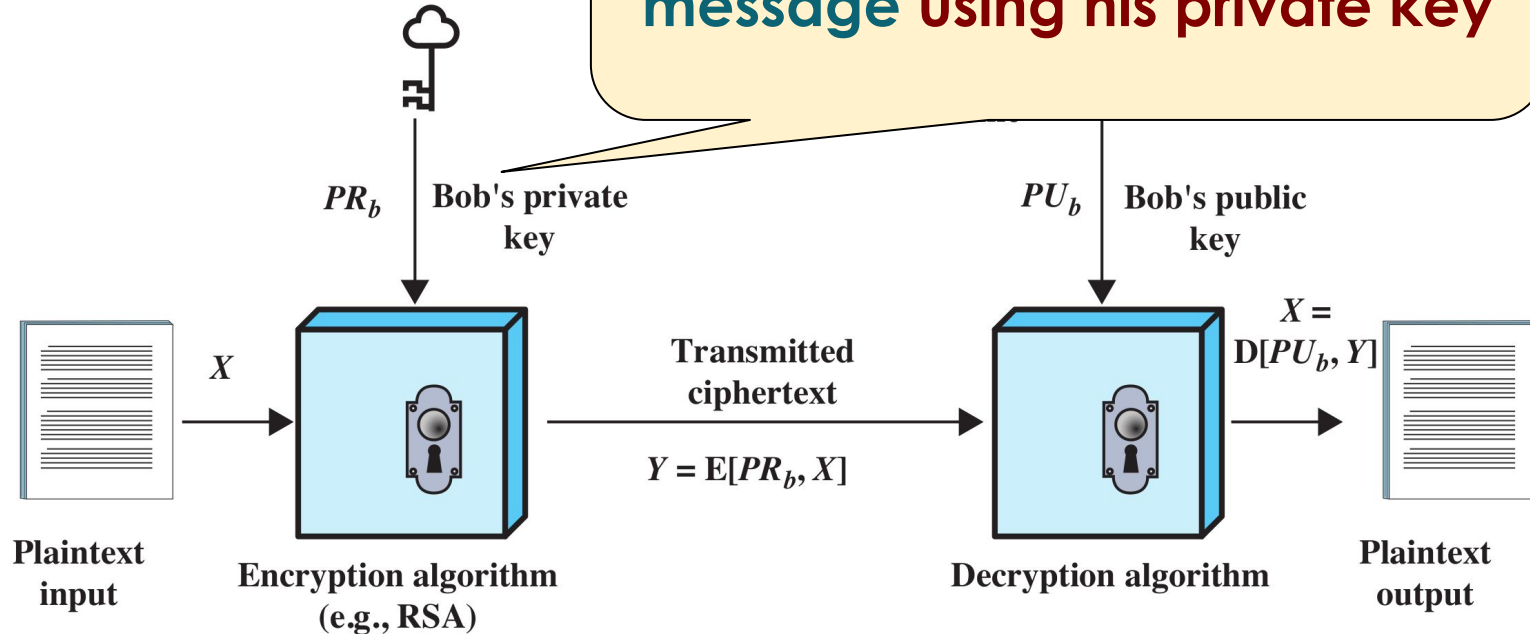


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A
L
I
C
E

In this case, Bob **encrypts** the
message using his private key

B
O
B



Public-Key Encryption



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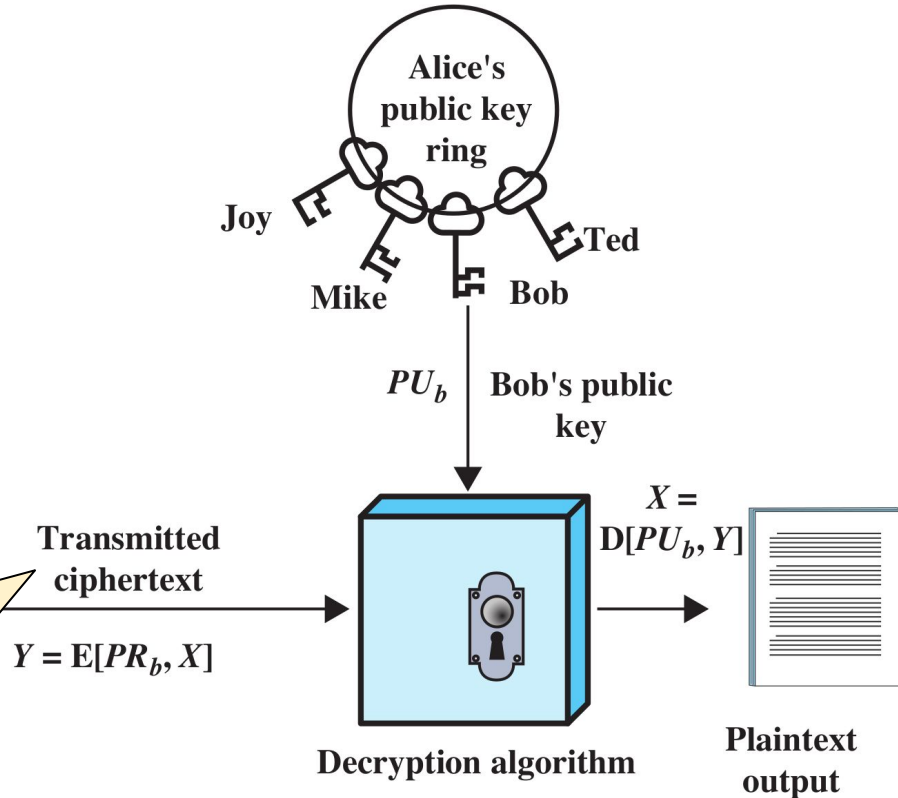
A
L
I
C
E

B

The **ciphertext** is
transmitted on the
Internet but it is **NOT**
confidential!



private
key



Public-

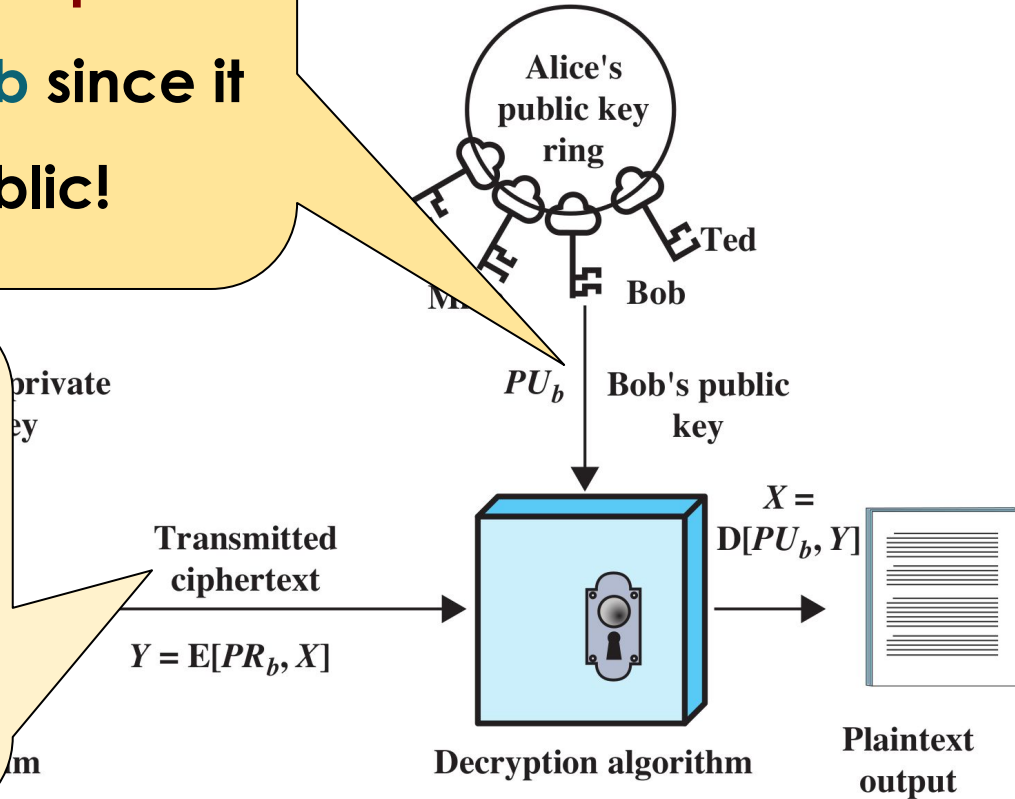
Because everyone can get the public key of Bob since it is public!



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A
L
I
C
E

The ciphertext is transmitted on the Internet but it is NOT confidential!



PU

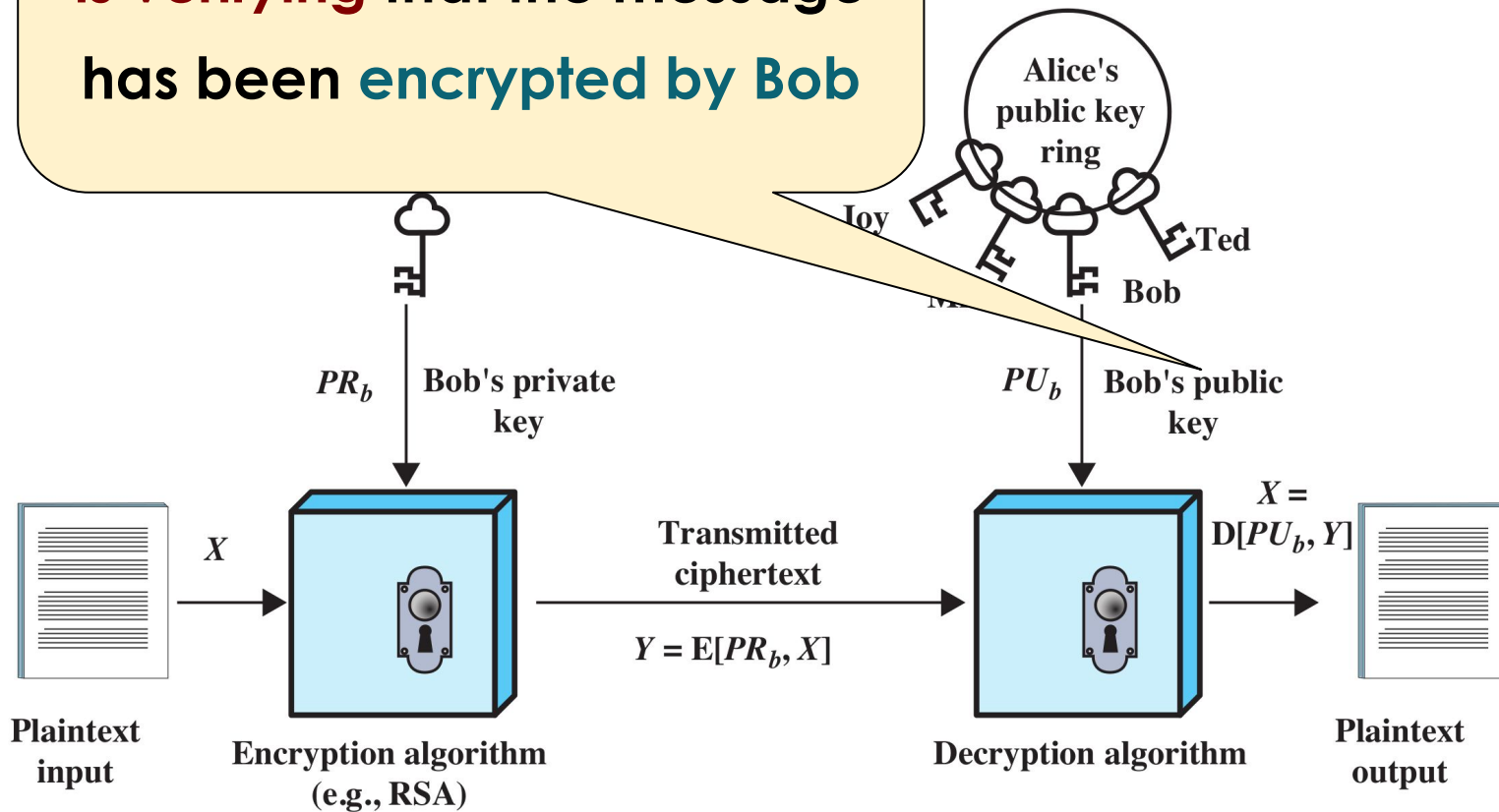
Using **Bob's public key**, **Alice** is **verifying** that the message has been **encrypted by Bob**



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A
L
I
C
E

B
O
B



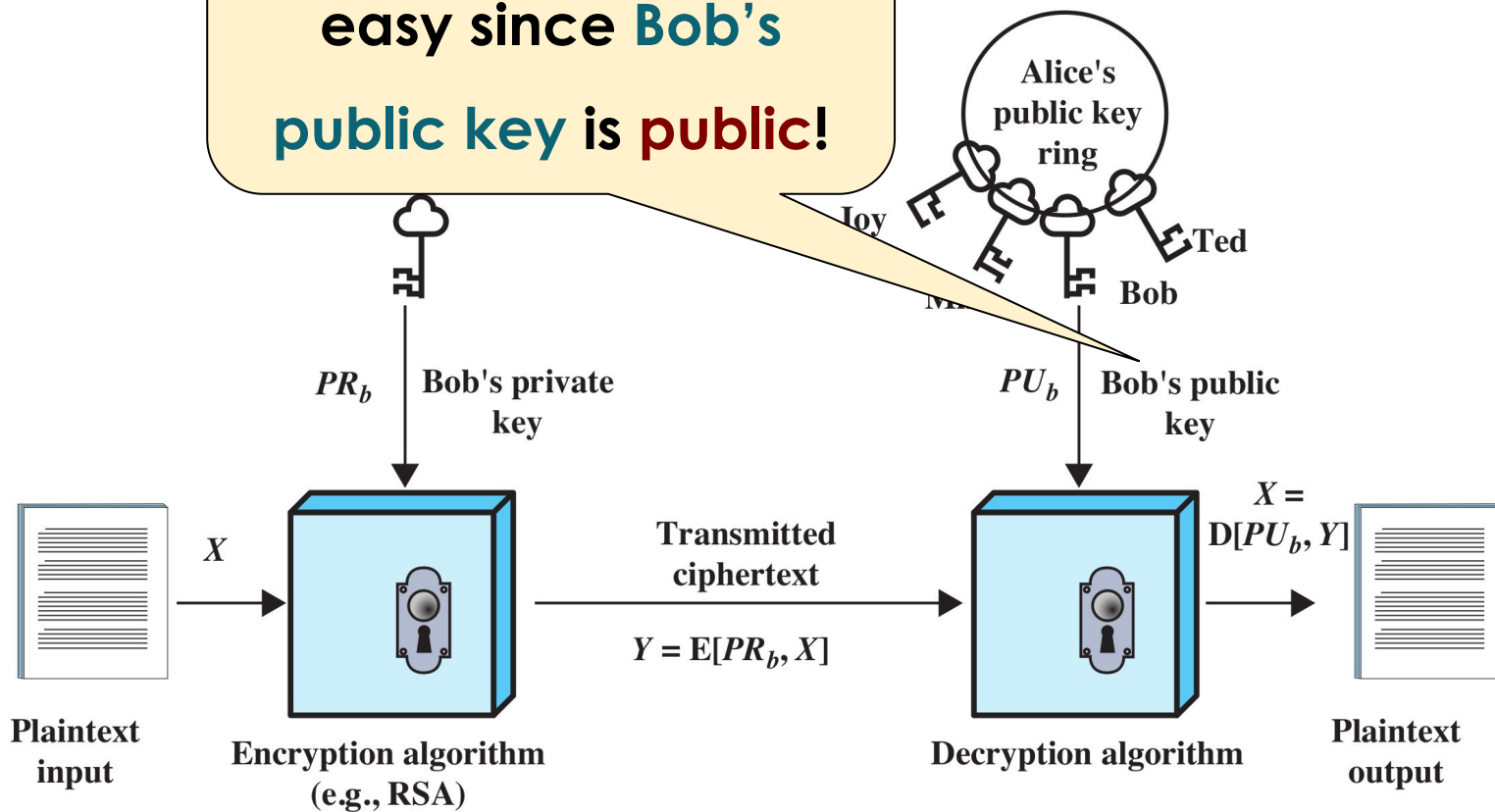
Public-key



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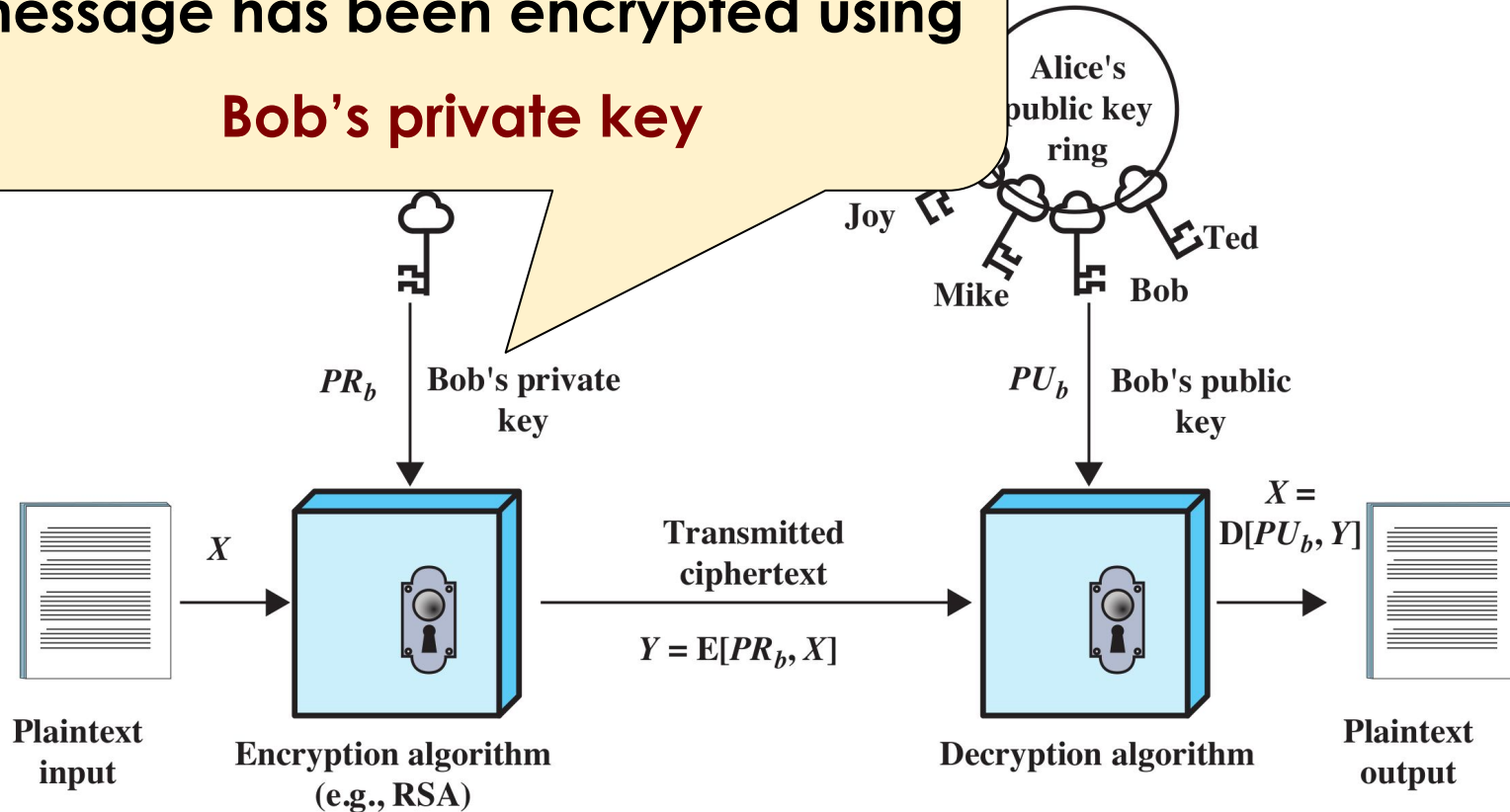
This **verification** is
easy since **Bob's**
public key is **public**!

A
L
I
C
E



B
O
B

The **verification** is **positive** only if the message has been encrypted using **Bob's private key**

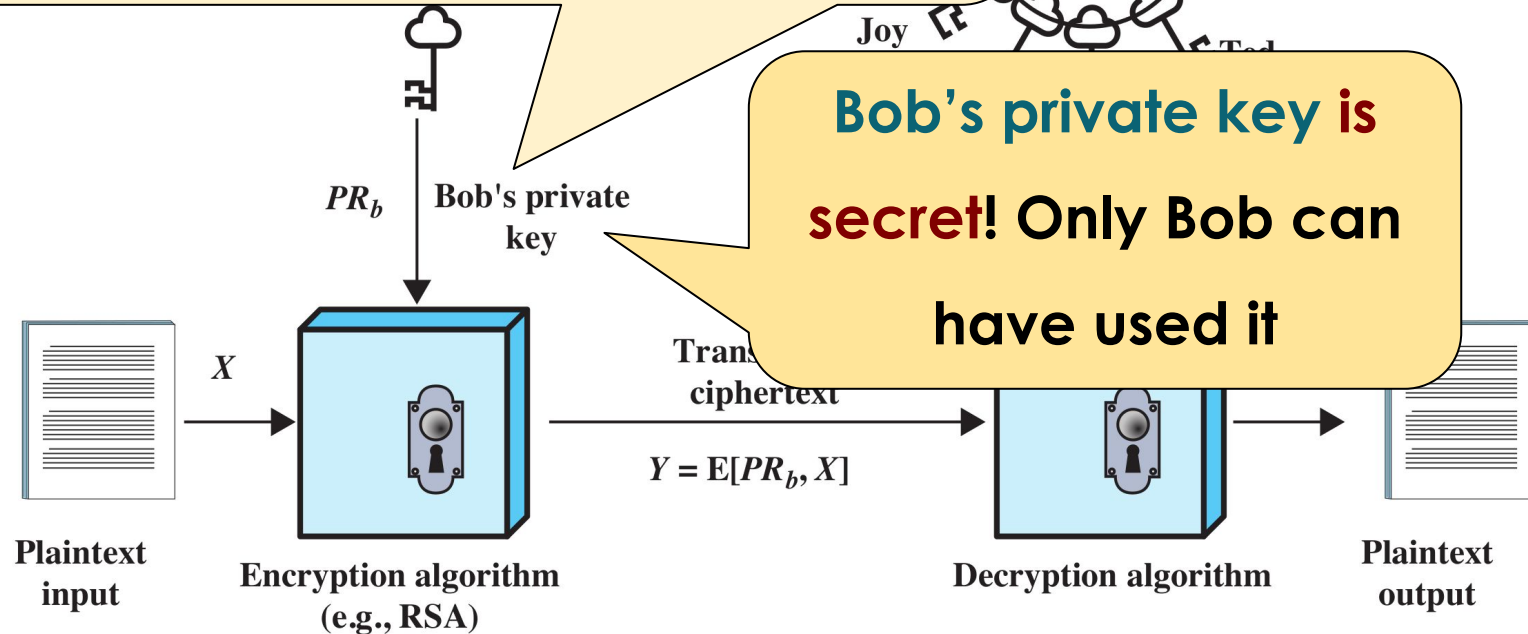
A
L
I
C
EB
O
B

The **verification** is **positive** only if the message has been encrypted using

Bob's private key

A
L
I
C
E

B
O
B



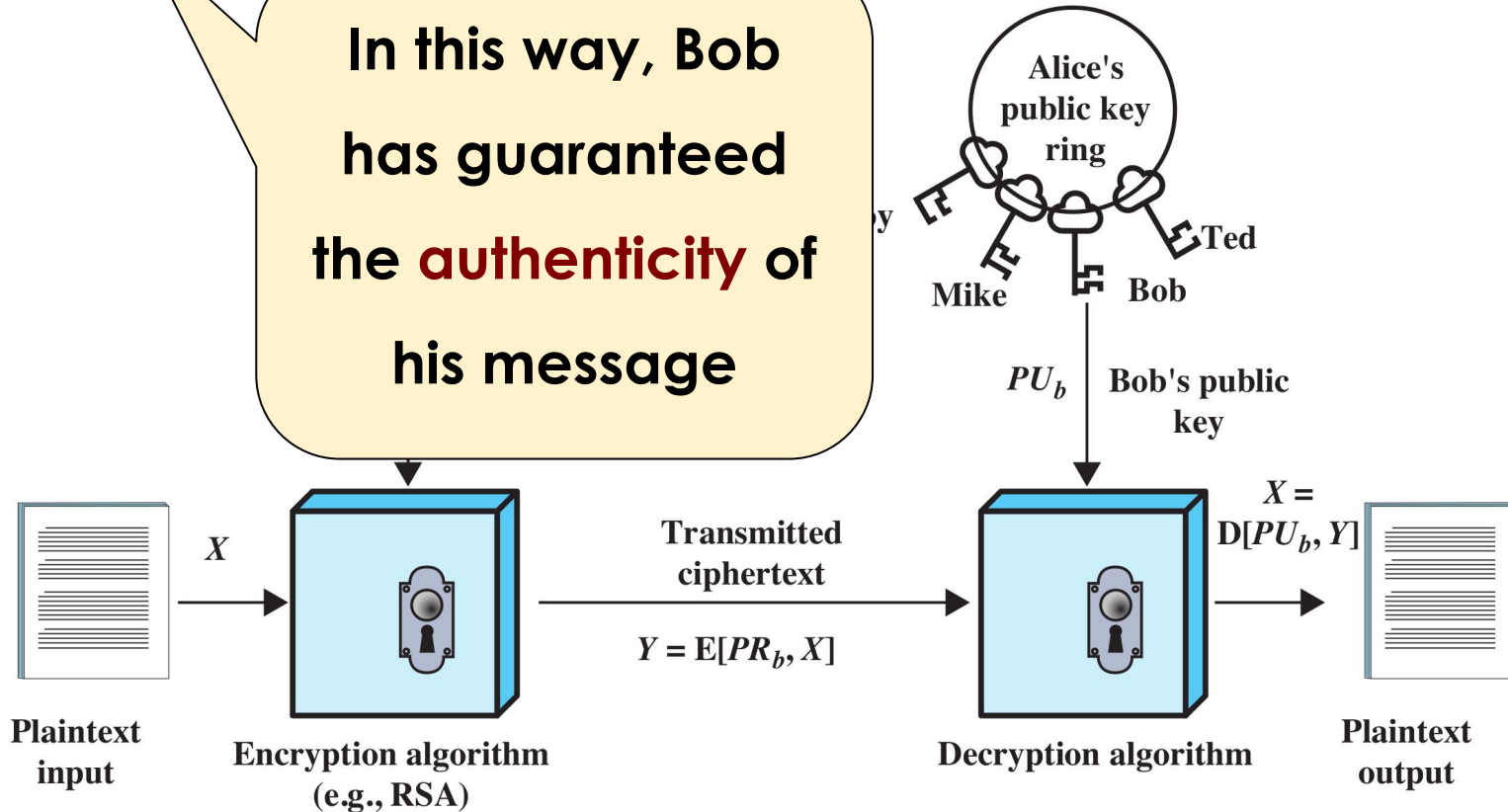
Public-Key Encryption



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In this way, Bob
has guaranteed
the **authenticity** of
his message

A
L
I
C
E



B
O
B

Public-Key Encryption

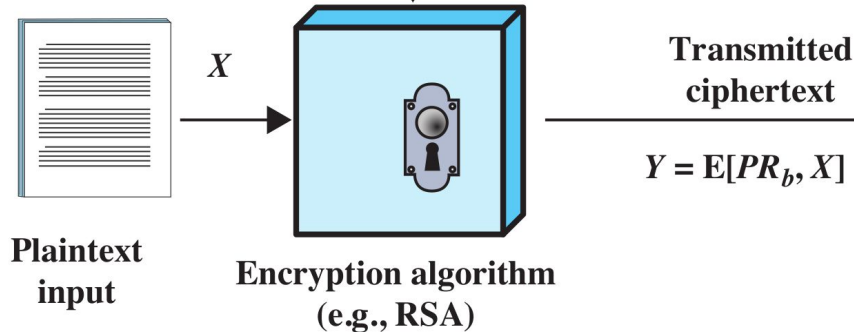


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A
L
I
C
E

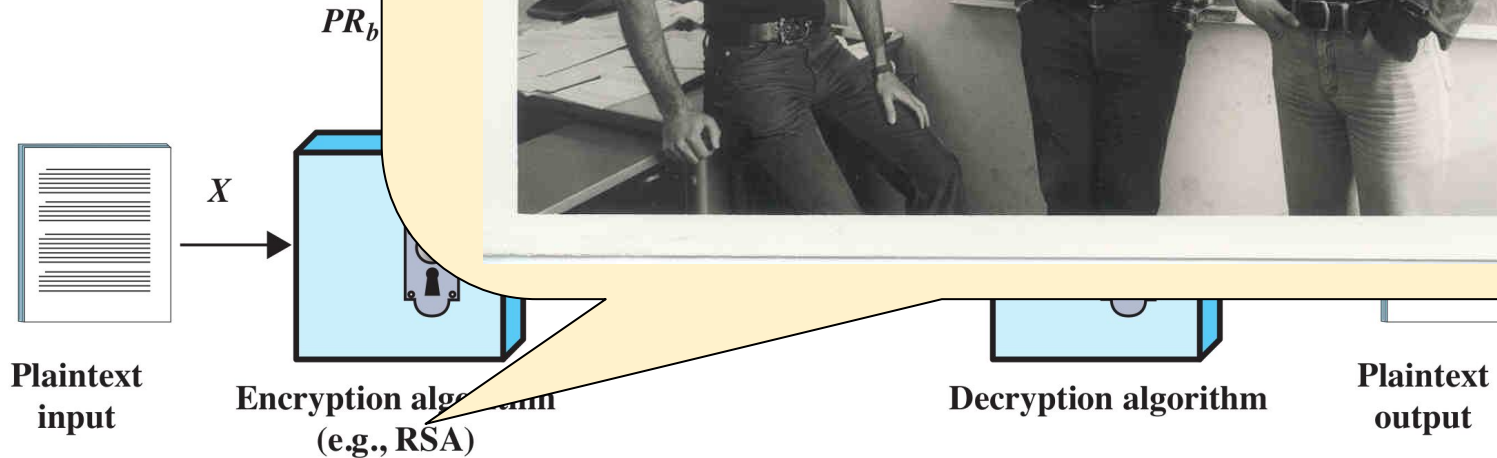
In this way, Bob
has guaranteed
the **authenticity** of
his message

This message is
really from Bob
and **not from an
impostor**



Public-Key E

B
O
B



E

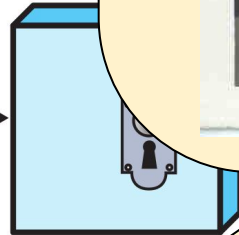
Public-Key E

B
O
B

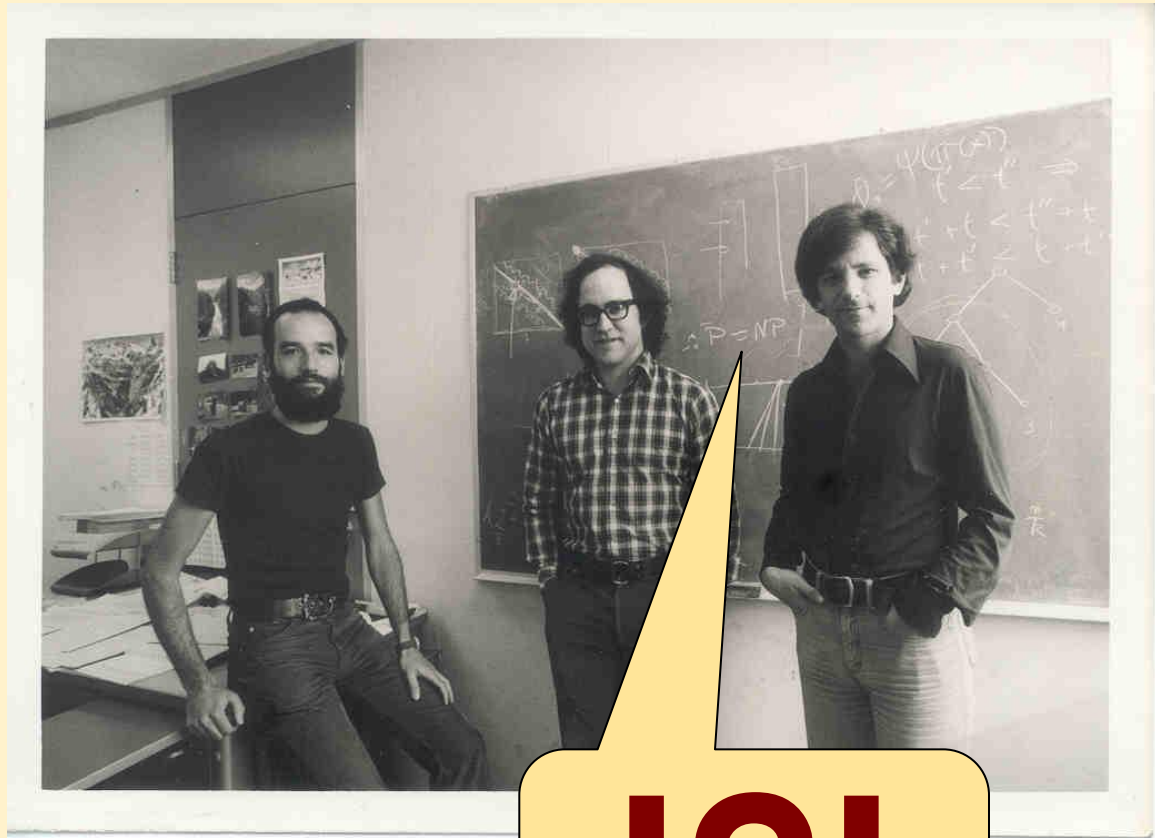


Plaintext
input

X



Encryption algorithm
(e.g., RSA)



!?!?

Dec

Text

E

Public-Key Encryption: requirements



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computationally **easy** to
create key pairs

useful if **either key** can
be used for **each role**

computationally **easy** for
sender knowing public key to
encrypt messages

computationally **infeasible**
for opponent to otherwise
recover original message



computationally **easy** for
receiver knowing private key to
decrypt ciphertext

computationally **infeasible** for opponent
to **determine private key** from **public key**

Public-Key Encryption: a big problem



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- The mechanism described before works very well but **there is still a problem**
- How is it possible to **guarantee** (i.e. **verify**) that the **Bob's public key is really the public key of Bob?**
- **For example:** *an impostor could impersonate Bob with the aim to publish a fake "Bob's public key"*

Public-Key Encryption: a big problem



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- The mechanism described before **is still a problem**
- How is it possible to **guarantee** (i.e. **really the public key of Bob?**)
- **For example:** *an impostor could impersonate Bob with the aim to publish a fake “Bob’s public key”*

And then?
With what consequences?



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GNU Privacy Guard



Usage examples of **GPG**

Encrypted emails?

Check it out!

T4



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Digital Signatures

Digital Signatures



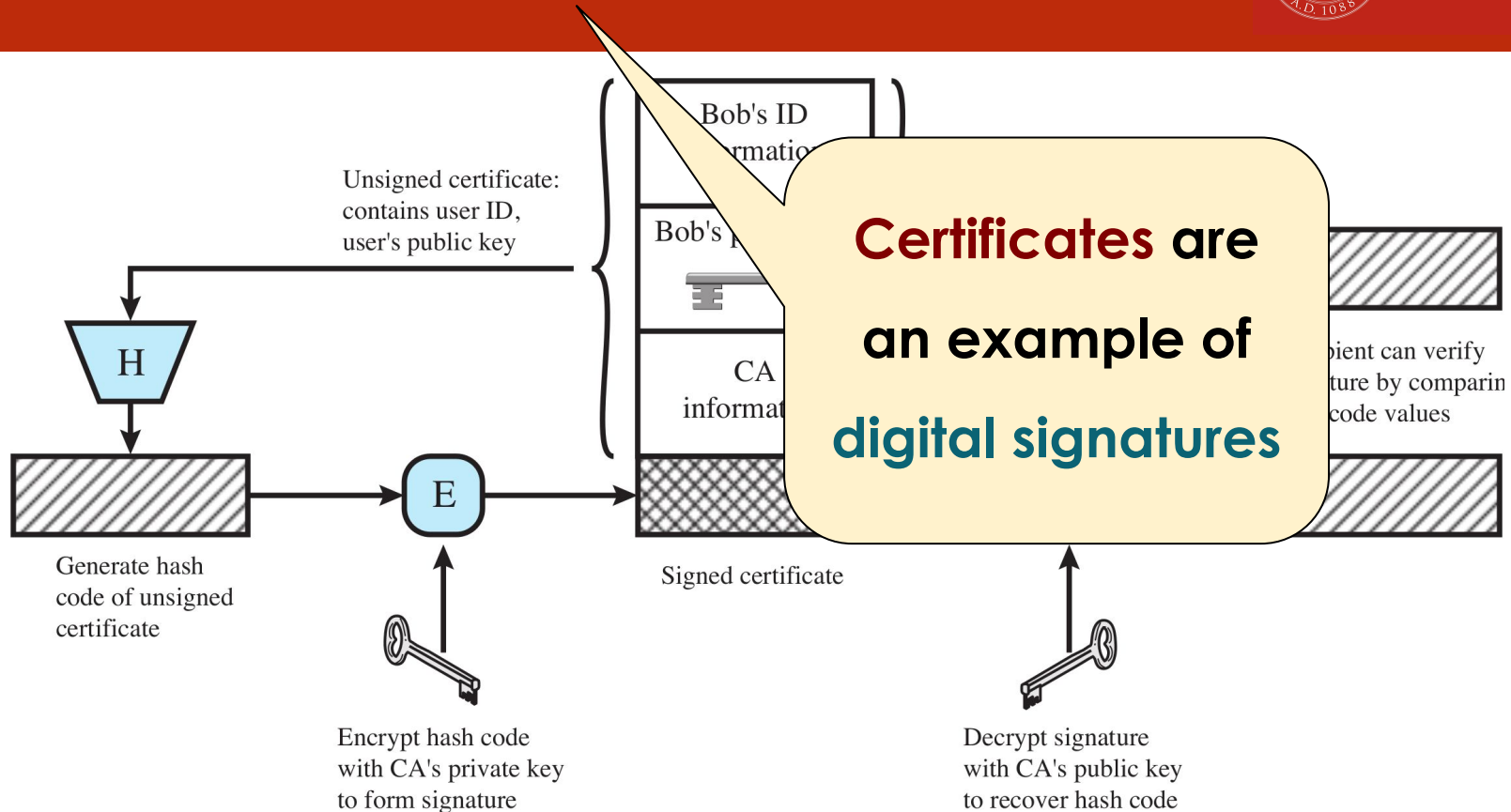
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- The **Digital Signature** is used for **authenticating** both **source** and **data integrity**
- Created by **encrypting hash code with private key**
- Does **not** provide **confidentiality**:
 - even in the case of **complete encryption** (e.g., not only the hash code)
 - the message is **safe from alteration** but **not eavesdropping**

Public-Key Certificates



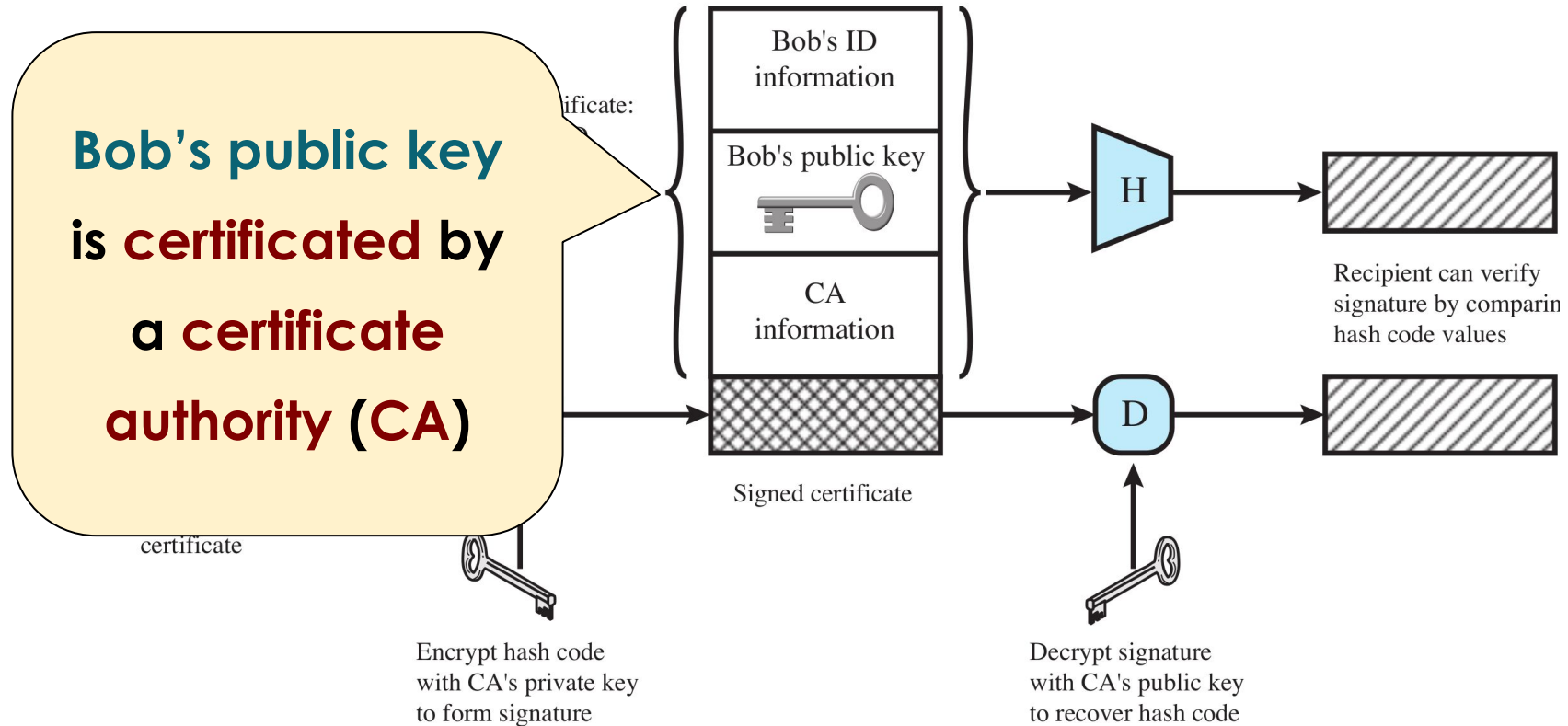
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Public-Key Certificates



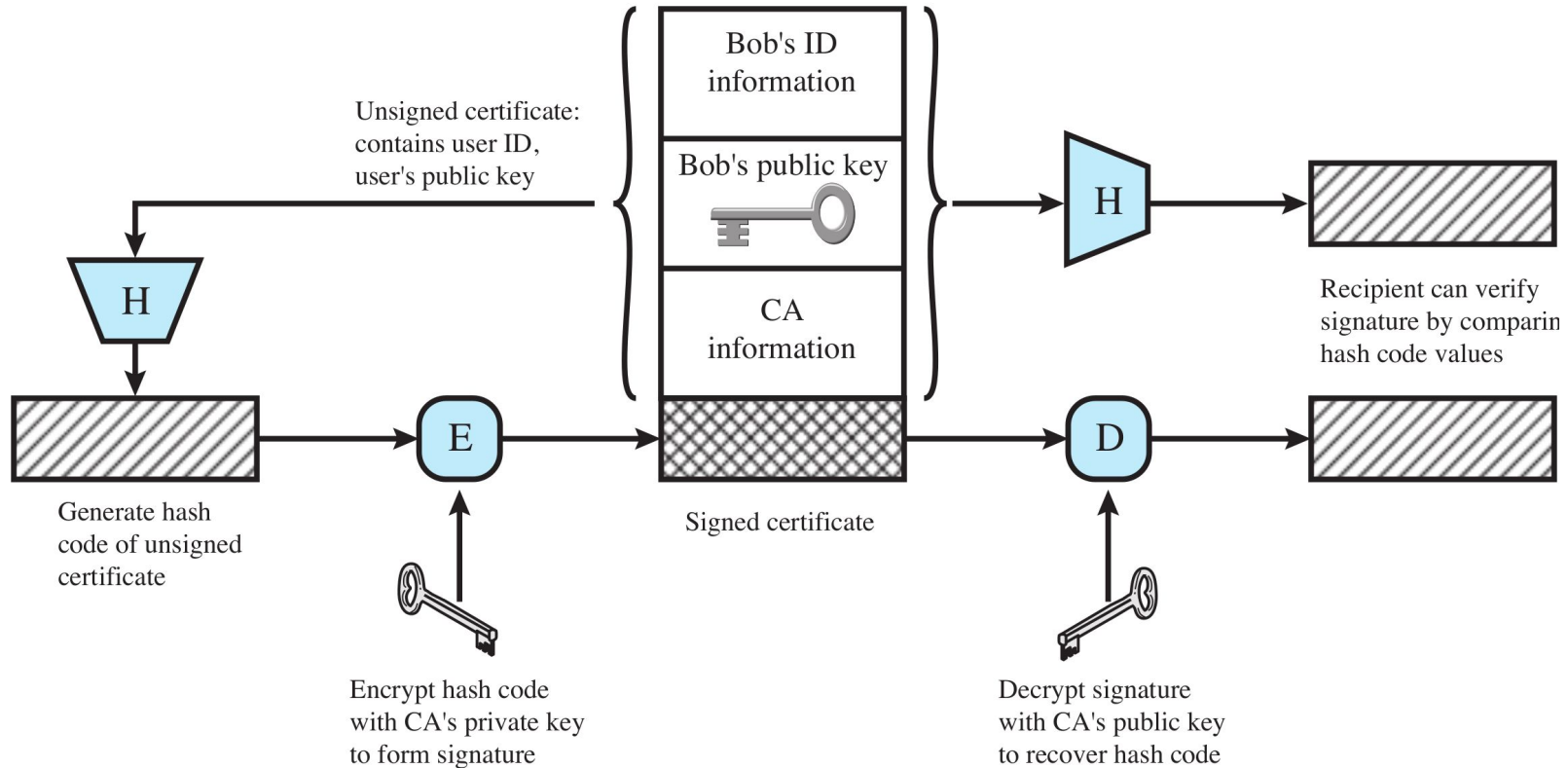
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Public-Key Certificates



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Public-Key Certificates

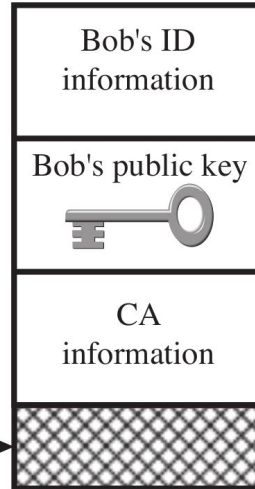


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**Step 1: an
unsigned
certificate is
prepared**

Generate hash
code of unsigned
certificate

Encrypt hash code
with CA's private key
to form signature



Signed certificate

Decrypt signature
with CA's public key
to recover hash code



Recipient can verify
signature by comparin
hash code values



Public-Key Certificates

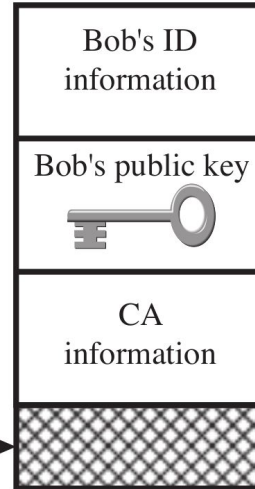


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**Step 1: an
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Encrypt hash code
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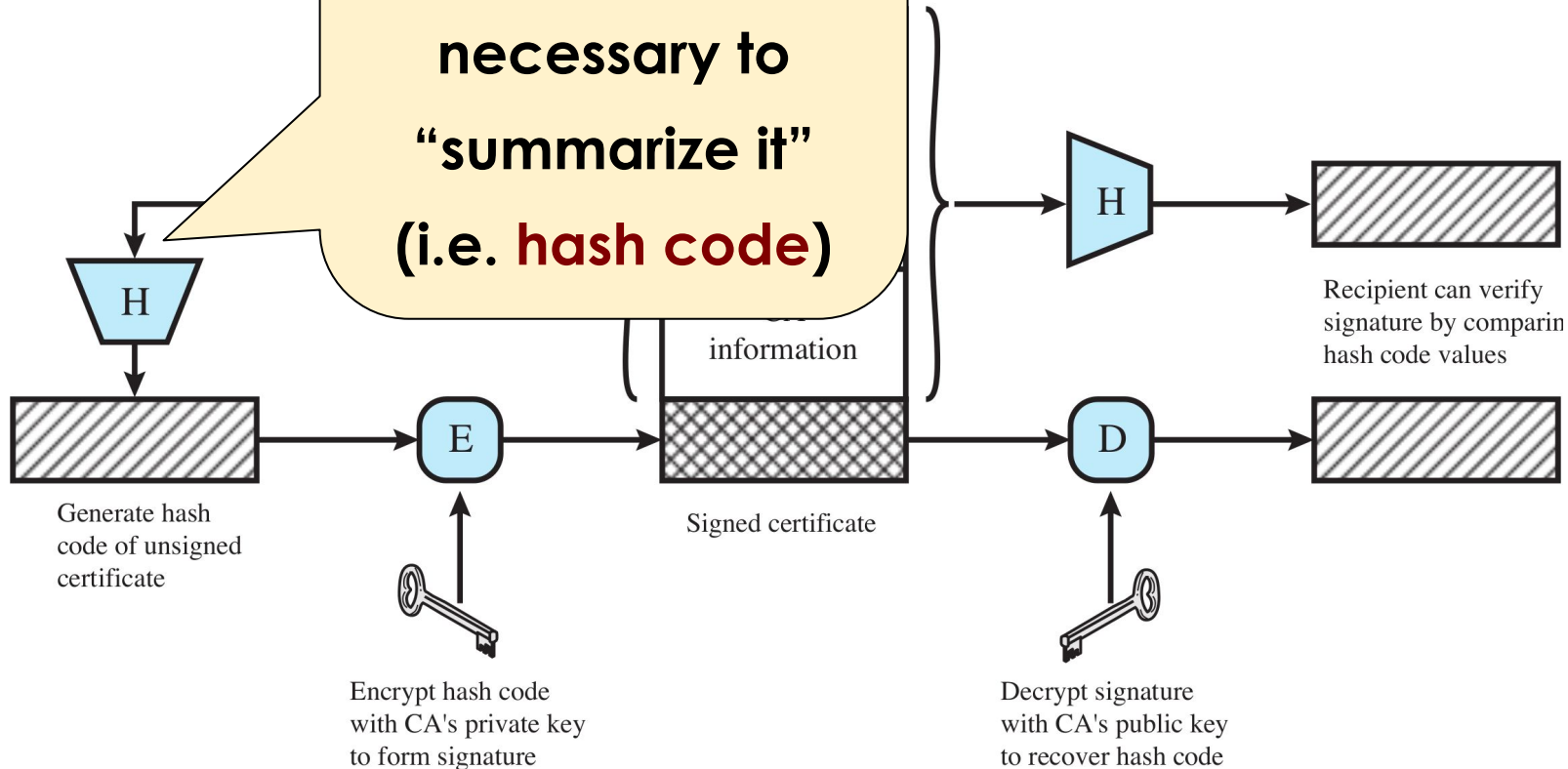


Signed certificate

Since it is **unsigned**
then it is easy to
forge the certificate
(create a fake one)

Decrypt signature
with CA's public key
to recover hash code

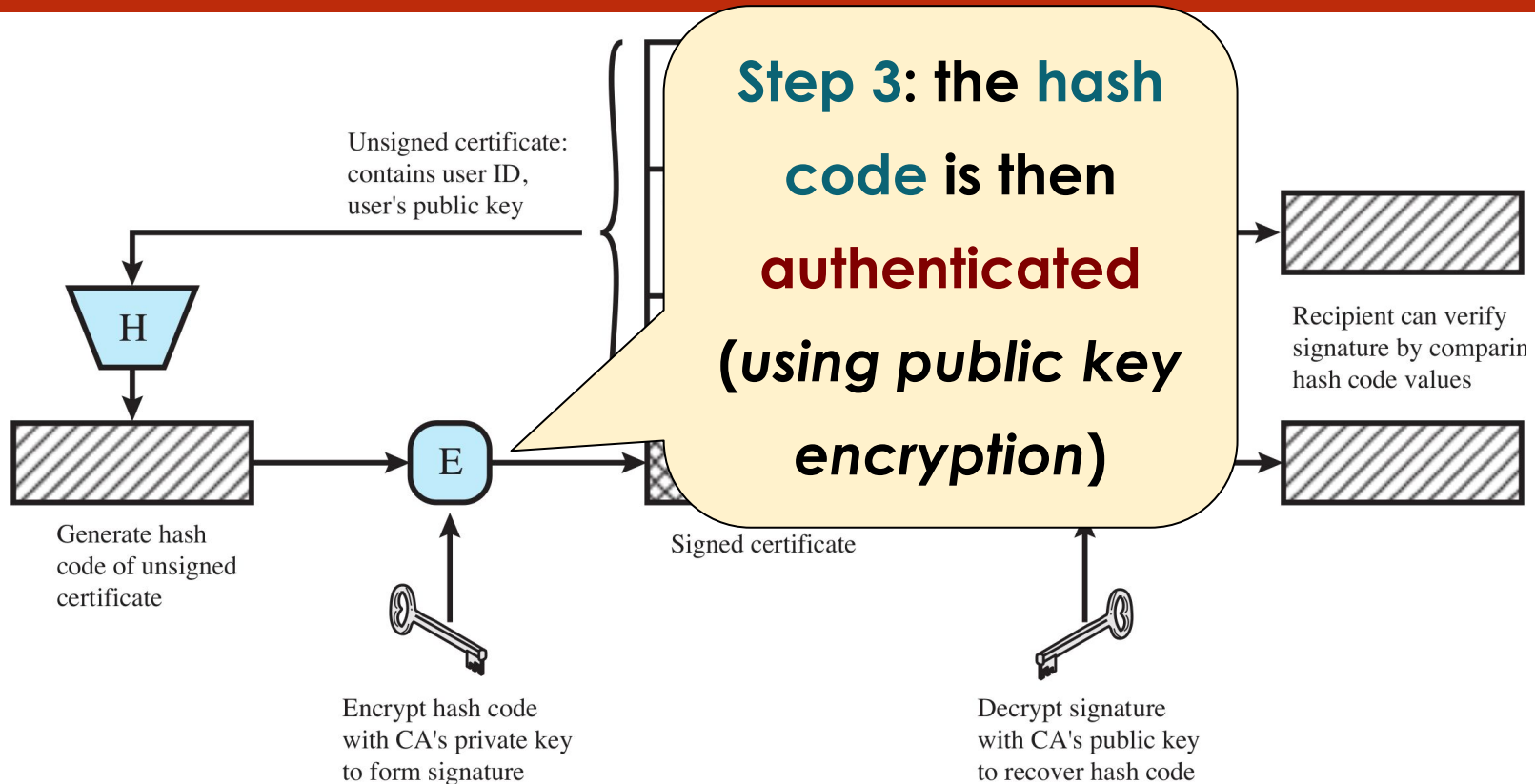
**Step 2: to sign the
certificate it is
necessary to
“summarize it”
(i.e. hash code)**



Public-Key Certificates



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Public-Key Certificates

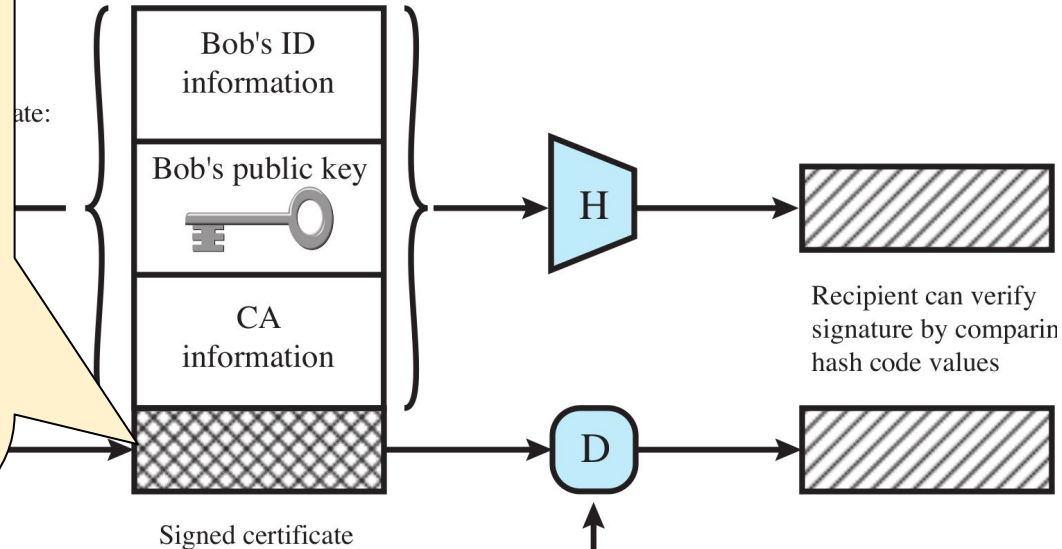


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**Step 4: ... and then
added to the
certificate that is
now **signed** (i.e. its
authenticity can be
verified)**

Generate hash
code of unsigned
certificate

Encrypt hash code
with CA's private key
to form signature

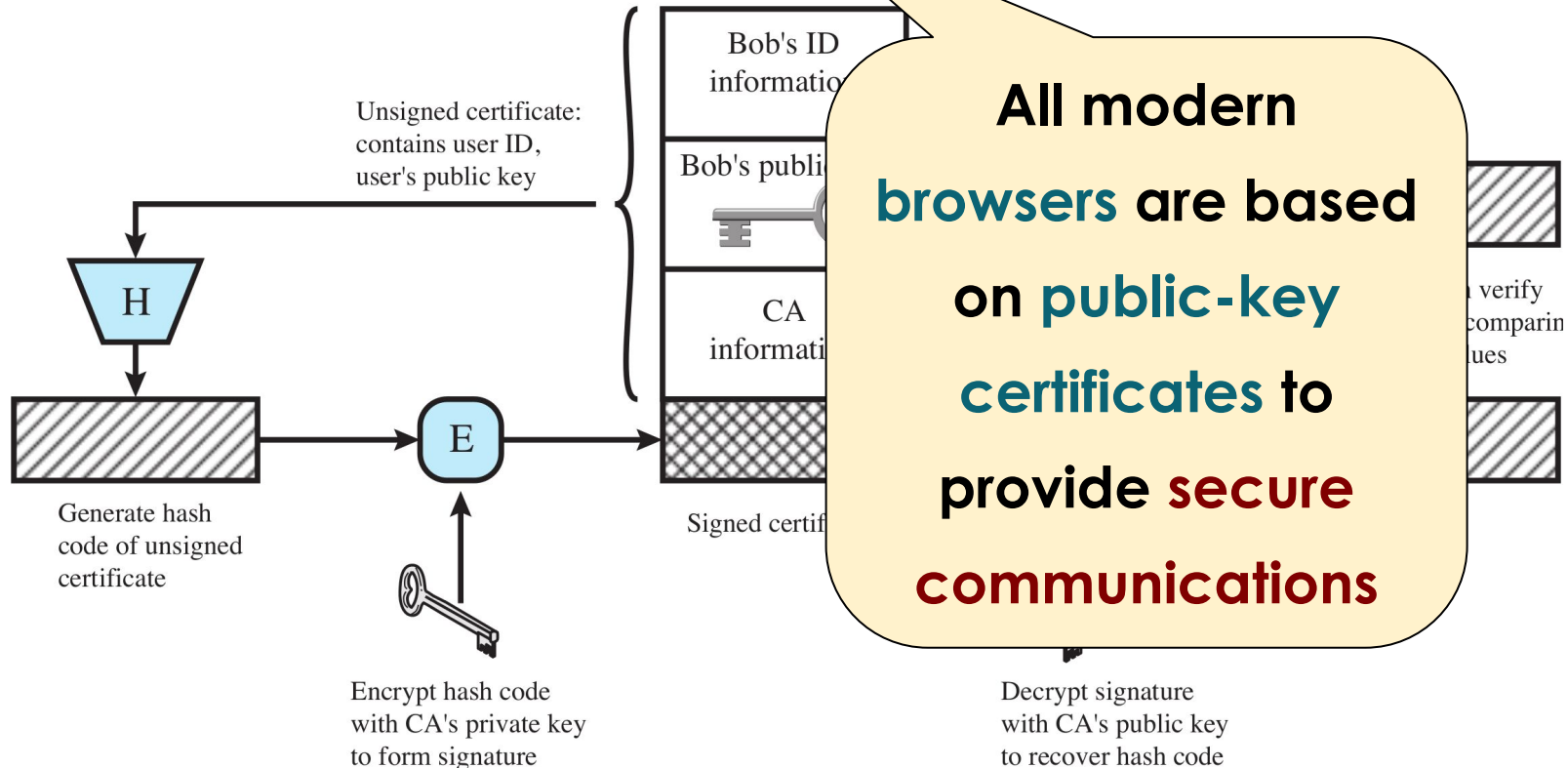


Decrypt signature
with CA's public key
to recover hash code

Public-Key Certificates



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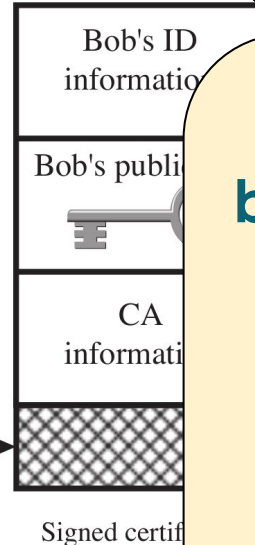
Public-Key Certificates



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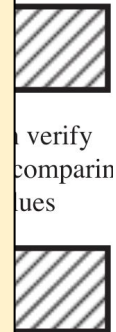
Using this method
the **browser can**
check that I'm
connected to the
proper website and
not to an impostor

with CA's private key
to form signature



All modern
browsers are based
on **public-key**
certificates to
provide **secure**
communications

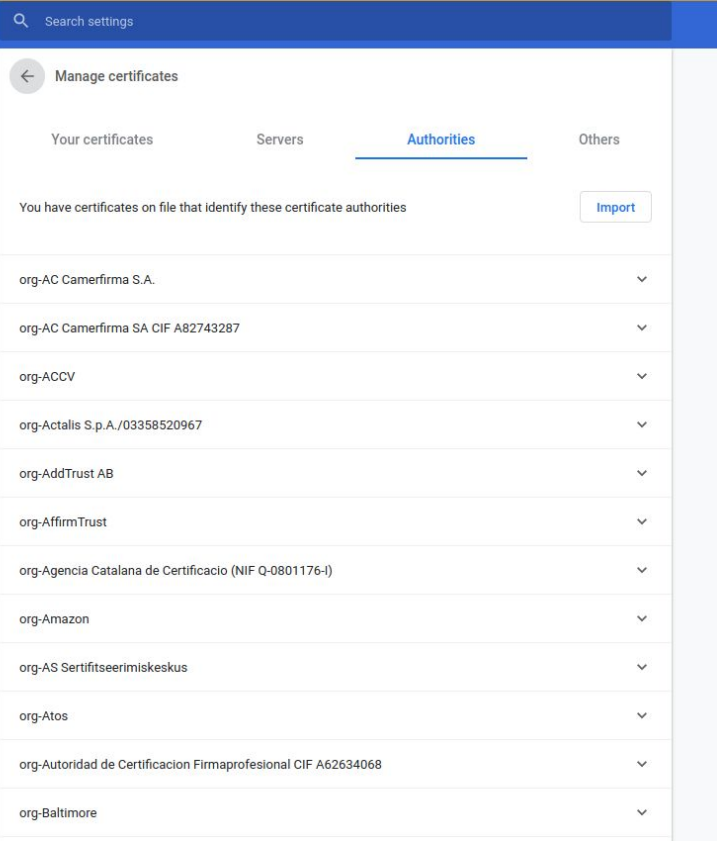
Decrypt signature
with CA's public key
to recover hash code



Web Browsers



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In Google Chrome (like in any other Web Browser) there is a **list of recognized certificates** from **trusted Certificate Authorities**

That are the **public keys** (of the **Certificate Authorities**) used to verify the certificates provided by the Web sites (*Amazon, Google...*)



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Pharming Attacks



Http



Https

Example of **pharming attack**
to **HTTP** and **HTTPS**

How does it works?

Check it out!

T5



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What's next?

Quantum Computing vs. Encryption



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- “A **quantum computer** is a computer that takes advantage of **quantum mechanical phenomena**” ([Wikipedia](#))
- Thanks to the “**quantum parallelism**” it can **speedup the execution of some specific computations (NOT all)**
- What is the expected effect of quantum computing on encryption?

Quantum Computing vs. Encryption



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- Most current **symmetric cryptographic algorithms** and **hash functions** are considered to be **relatively secure**
- In case, it is possible to switch to **longer keys** and **hashes**
- This is **not true** for current **public-key algorithms**
- A new generation of **post-quantum cryptography** algorithms is being developed, standardized and adopted in software (e.g. browsers)

Quantum Computing vs. Encryption



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Barış Ege

PRINCIPAL SECURITY ANALYST,
KEYSIGHT DEVICE SECURITY
TESTING



#Cybersecurity

#Quantum Computer

#Quantum Computing

NETWORK VISIBILITY + SECURITY

Security Highlight: China's Quantum Leap, and Why RSA Isn't at Risk (Yet)

2024-10-28 | ⌚ 4 min read

Recent headlines suggested that Chinese scientists have the potential to break military-grade RSA encryption using quantum computers, causing concerns over online data security. However, upon closer inspection, these claims do not yet signal the end of modern encryption. Here's what you need to know.

Understanding the breakthrough

The Chinese Journal of Computers recently **published a paper** in which researchers from Shanghai University factored a 22-bit integer using D-Wave's quantum annealing machine. The research showcases how quantum annealing can be leveraged to transform cryptographic attacks into solvable combinatorial optimization problems.

The team factored a 22-bit number, a far cry from the 2048-bit keys currently used for secure encryption. The difference between a 22-bit and a 2048-bit number is exponential, and this study, while a step forward, doesn't yet threaten RSA encryption as we know it.

Quantum Computing vs. Encryption



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Key-size:
22 vs. 2048 bits

NETWORK VISIBILITY + SECURITY

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#Quantum Computer

#Quantum Computing

Quantum Computing vs. Encryption



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ars TECHNICA

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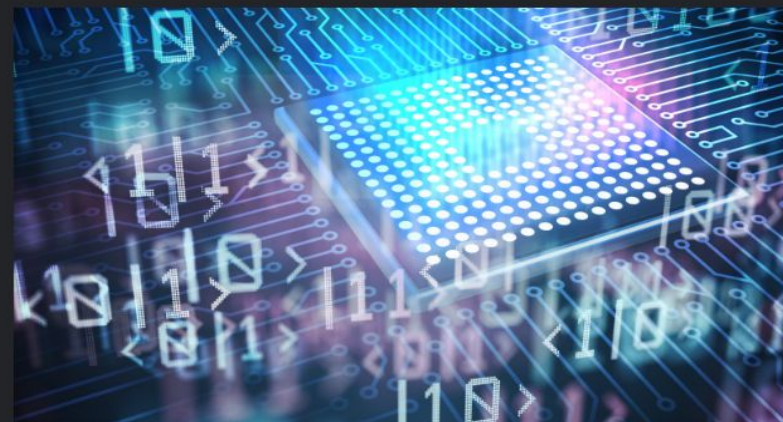


CRYPTOGRAPHICALLY RELEVANT QUANTUM COMPUTING

Quantum computers need vastly fewer resources than thought to break vital encryption

No, the sky isn't falling, but Q Day *is* coming, and it won't be as expensive as thought.

DAN GOODIN – MAR 31, 2026 8:25 PM | 39



➔ Credit: vital



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Digital Envelopes

Digital Envelopes



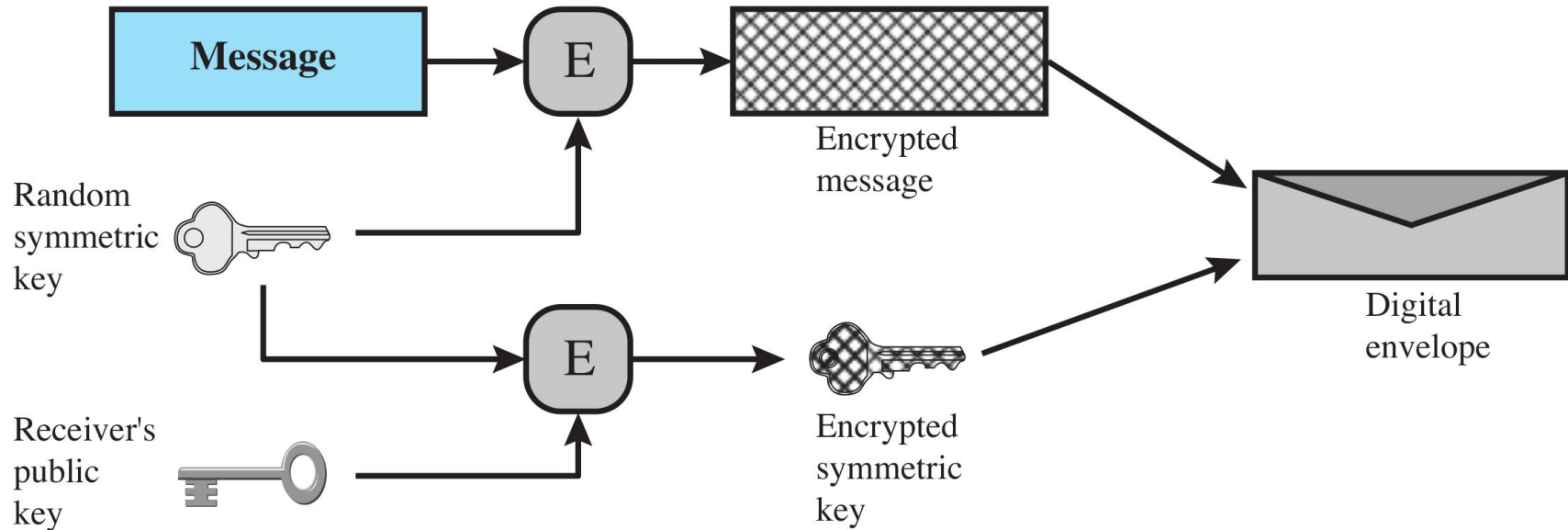
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- They are used to **protect a message** without needing to arrange beforehand, for sender and receiver, to have the **same secret key**
- It equates to the same thing as a **sealed envelope** containing an **unsigned letter**
- They have some interesting characteristics

Digital Envelopes: creation



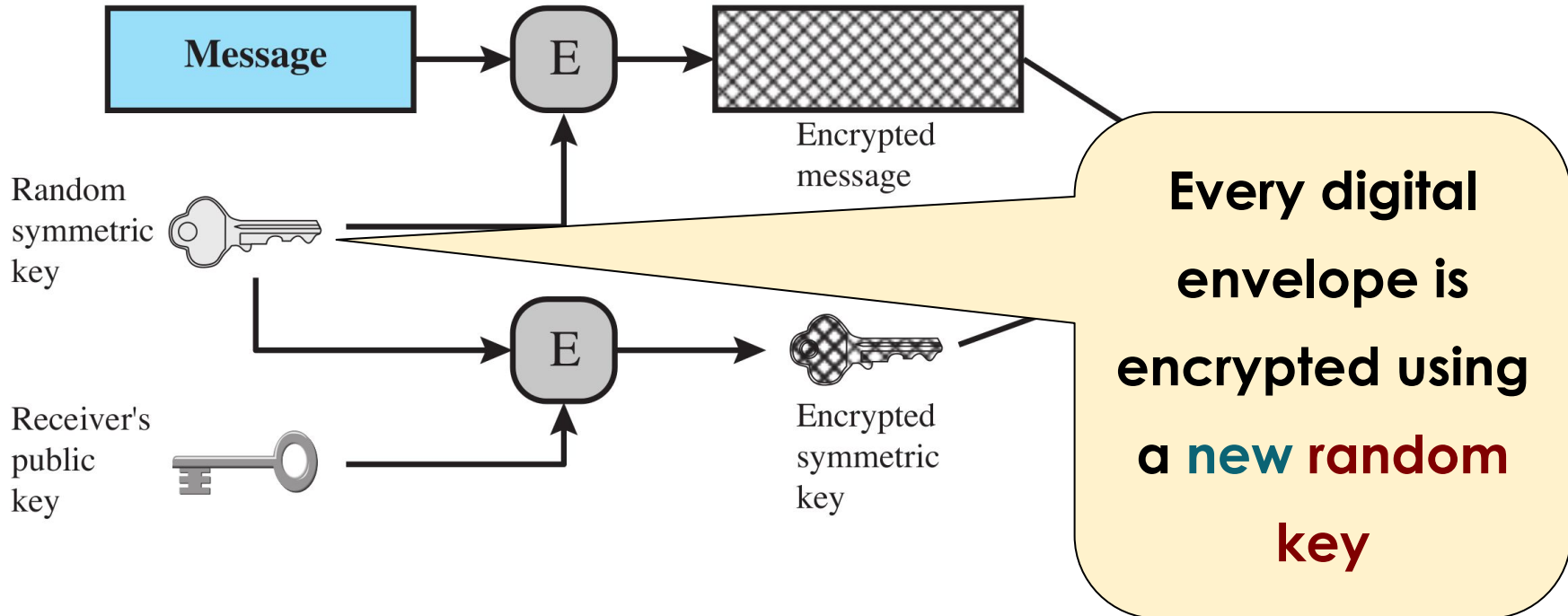
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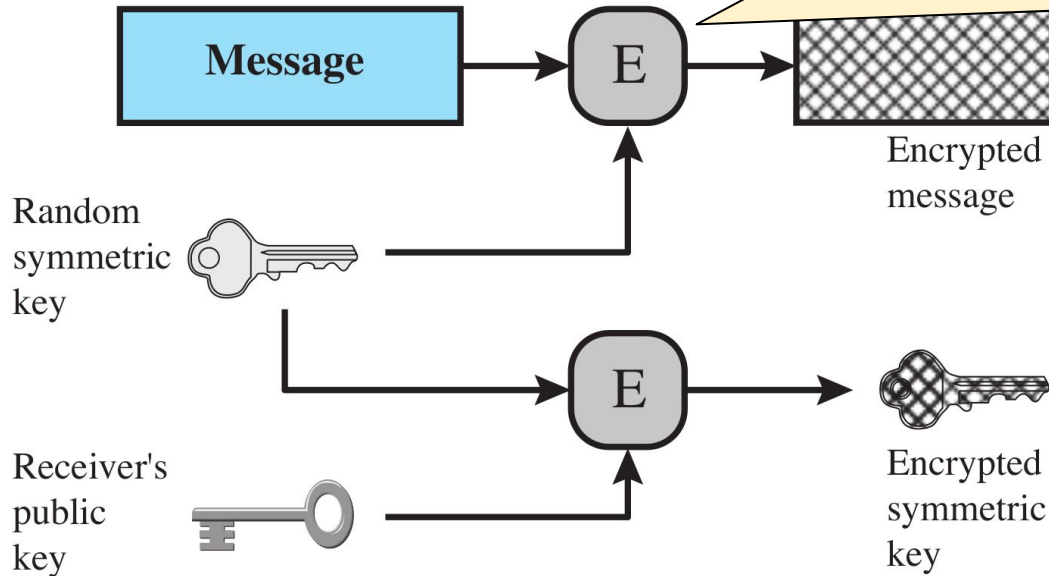
Digital Envelopes: creation



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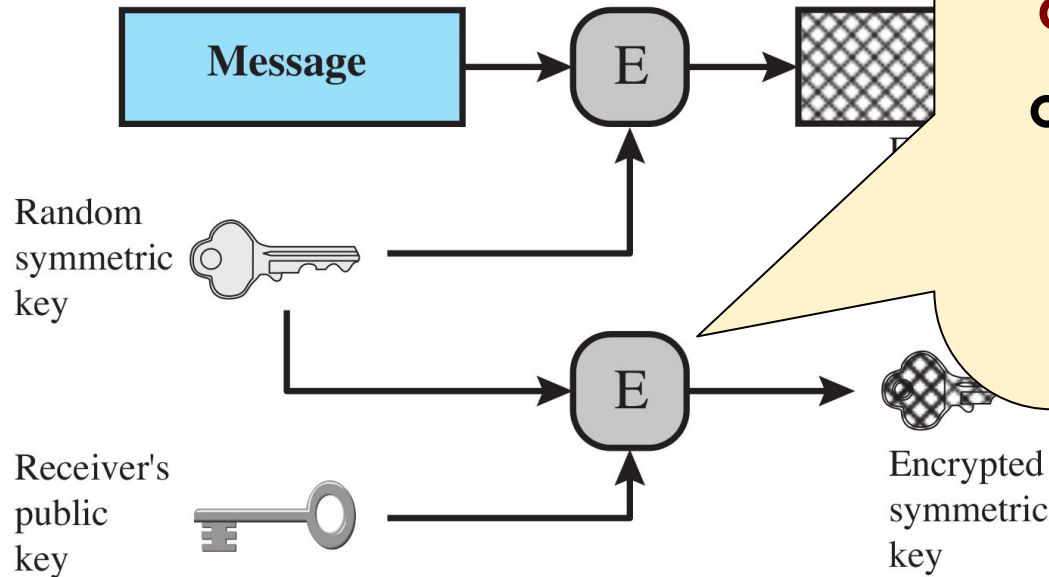


Digital Envelopes: creation



The digital envelope is encrypted using a **symmetric algorithm** that is **much faster** than **public key encryption**

Digital Envelopes: creation

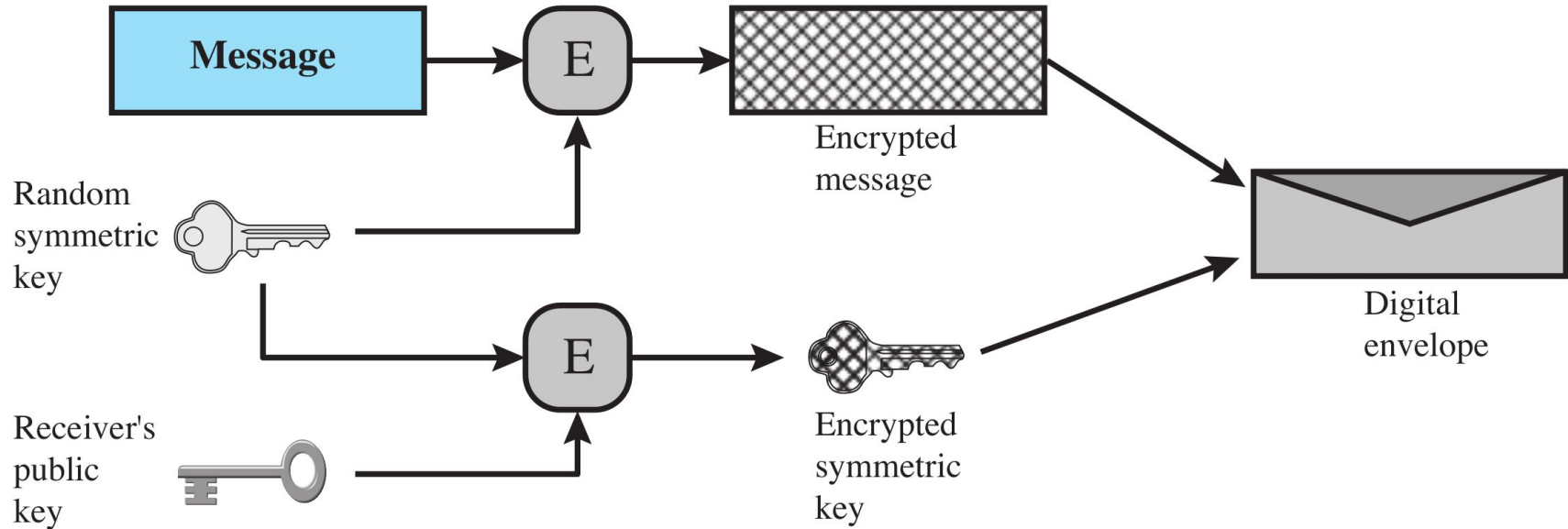


The **public key encryption** (that is **quite slow**) is used only for **encrypting the random symmetric key**

Digital Envelopes: creation



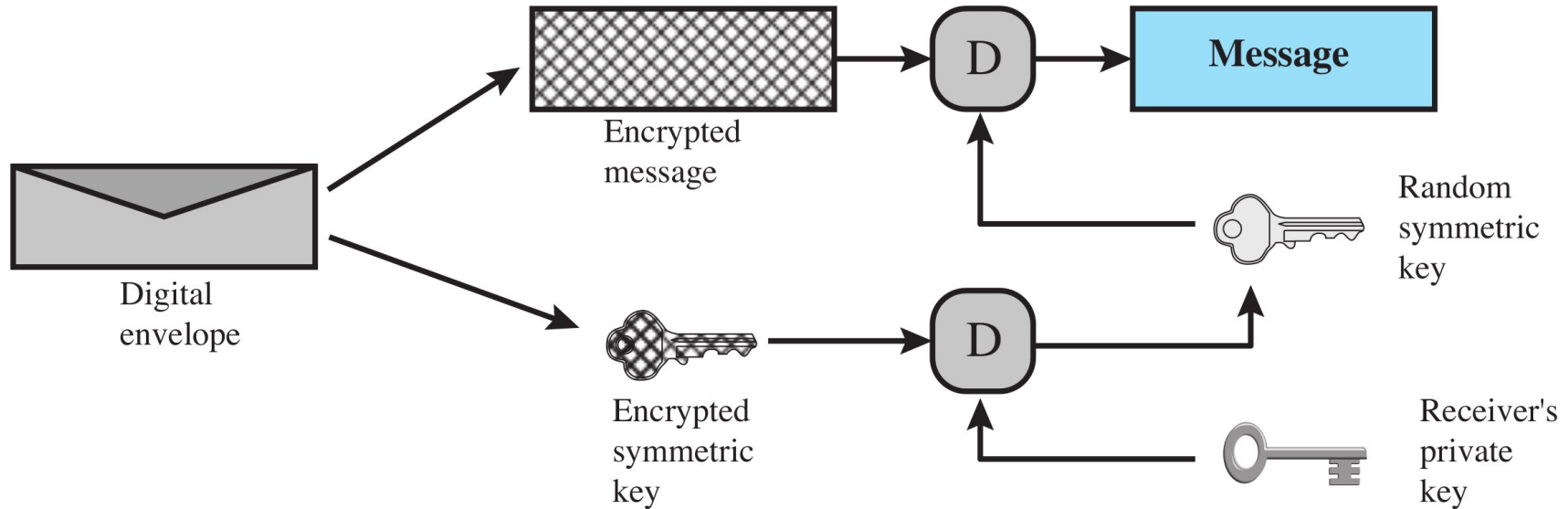
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Digital Envelopes: opening



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Random Numbers

Random numbers (**generation**)



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- **Random numbers** are required by many different tasks:
 - **generation of keys** for **public-key algorithms**
 - **stream key** for **symmetric stream cipher**
 - **symmetric key** to use as a **temporary session key** or to create a **digital envelope**
 - **handshaking** to **prevent replay attacks**
 - **session key** (i.e., randomly generated symmetric key)

Random numbers (**generation**)



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- **Random numbers** are required by many different tasks:
 - **generation of keys** for **public-key**
 - **stream key** for **symmetric stream**
 - **symmetric key** to use as a **temporary**
digital envelope
 - **handshaking** to **prevent replay attacks**
 - **session key** (i.e., randomly generated symmetric key)

**An adversary is
repeating a previously
captured user response**

Random numbers (**requirements**)



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- **Randomness criteria:**
 - **uniform distribution:** frequency of occurrence of each of the numbers should be **approximately the same**
 - **independence:** **no one value** in the sequence can be **inferred from the others**

Random numbers (requirements)



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- **Randomness criteria:**
 - **uniform distribution:** frequency of occurrence of each of the numbers should be **approximately the same**
 - **independence:** no one value in the sequence can be **inferred from the others**

Unpredictability

Random numbers (requirements)



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Each number is
statistically
independent of other
numbers in the
sequence

quency

proxima

value i

Opponent should **not**
be able to predict
future elements of the
sequence on the basis
of earlier elements

Unpredictability

Randomness vs. Pseudo-randomness



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- **Cryptographic applications** typically use **algorithms** for **random number generation**: algorithms are **deterministic** and therefore produce sequences of numbers that are **not statistically random**
- **Pseudo-random numbers** are:
 - sequences produced that **satisfy statistical randomness tests**
 - likely to be **predictable**

Randomness **vs.** Pseudo-randomness



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- **True Random Number Generator (TRNG):**
 - uses a **nondeterministic source** to produce **randomness**
 - most operate by measuring **unpredictable natural processes**
(e.g., radiation, gas discharge, leaky capacitors)
 - **increasingly provided on modern processors**

Randomness vs. Pseudo-randomness



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- **True Random Number Generator (TRNG):**
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(e.g., radiation, gas discharge)
 - increasingly provided on **modern processors**

Backdoors?

Randomness vs. Pseudorandomness

- **True Random Number Generators**

- uses a **nondeterministic** process
- most operate by measuring a physical phenomenon (e.g., radiation, gas discharge, thermal noise, capacitors)
- increasingly provided on **modern processors**



Backdoors? It is possible but carefully consider your **threat model**

Do you **trust your CPU?**

- increasingly provided on **modern processors**



Randomness (NOT MANDATORY)



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- If you are interested in **how random numbers generation is managed** in the new versions of the **Linux kernel**: “Uniting the Linux random-number devices”
- If you are interested in a (potential) **backdoor in an encryption standard** (not related to random numbers generation): “Dual Elliptic Curve Deterministic Random Bit Generator”



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CyberSecurity

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Attributions and Notes



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- Most of the slides are from: “**Computer Security: Principles and Practice**”, 4th edition, W. Stallings and Lawrie Brown
- Some others are from: Dr. **Mehmet H. Gunes** (Department of Computer Science & Engineering - University of Nevada, Reno)
- Some others are from: **Mario Cagalj**, Ph.D. (Faculty of Electrical Engineering, Mechanical Engineering and Naval Architecture (FESB), University of Split)
- I wish to **thank all of them**, **any errors that remain are my sole responsibility**